When the Neutrinos Leave: Decoupling History and the Threshold for Primordial Black Hole Formation

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AI-assisted. The ideas and research direction are the author's; AI tools (Claude, Anthropic) were used for literature search, computation, drafting and typesetting. The numbers come from the scripts, run logs and outputs printed in Appendix B. The simulations are runs of S. McCormick's public code on a cloud computer, not experiments.

Abstract

McCormick has shown that neutrinos which stream freely out of a collapsing overdensity raise the threshold for primordial black hole formation, by about $0.235\,R_\nu$ in the linear compaction, and that the effect fades when the neutrinos decouple gradually. This paper asks what sets that fading. First, the neutrino collision operator is read exactly as a cut between what is kept and what streams. All of its non-normality, $\frac{1}{8}\|[L, L^*]\|^2 = \kappa^2 k^2/3$, lies on the single link between density and flux. With separate rates for flux and shear it is $k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$, and the weight across the cut equals the sound speed, $\sqrt{E_B} = k/\sqrt{3}$. Second, McCormick's own code is run on five cases outside his paper, with every prediction and kill line written down before the runs. A broad profile that collapses near $\epsilon \approx 7$–9 keeps the same share of the neutrino effect as a Gaussian that collapses near $\epsilon \approx 4$, or less ($G = 0.181$ and 0.045 against 0.190 and 0.063), so the collapse time alone does not set the clock. A steeper decoupling history, $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^6$, gives $G = 0.648$, 0.317 and 0.154 at $\epsilon_{\rm dec} = 1$, 2 and 3, against 0.683, 0.431 and about 0.30 for the standard history: how the neutrinos leave matters, not only when. The steep-history thresholds are unchanged by a tenfold higher collision-rate cap and by a finer grid. One reading survives every run: the memory of the kept sector, the population the density keeps under the decoupling operator, read mostly at the first compaction maximum. With one weight fitted to McCormick's curve and then frozen, it predicted 0.663, 0.246 and 0.118 before the runs, all within the kill line of 0.08; a history-blind rule fails at $\epsilon_{\rm dec} = 3$ by 0.146. Third, the memory has a location. Two neutrino states with the same density, flux and anisotropic stress, differing only in finer angular structure, give later density, flux and stress that differ by several per cent, when the difference is set at or after decoupling; set deep in the coupled era it is erased, by the factor $\exp[-\epsilon_{\rm dec}^n/(n\tau_s^n)]$. An exact block elimination puts the whole effect in one hidden-source term, and the kept population obeys $(\dot{A})^2 \le 4E_B A(1-A)$. A closure on density, flux and stress cannot carry this memory. The algebraic identities, the counterexample and the verdict arithmetic are written in Lean for machine checking (Appendix C). Four further tests are stated for the $e^\pm$ annihilation epoch and for collision terms beyond relaxation time.

1 The question

Primordial black holes heavier than about $10^6\,M_\odot$ form while the neutrinos, which carry 41–49% of the radiation, stop scattering and stream freely. McCormick evolved these neutrinos with their

angle-resolved transport equation inside spherical general relativity and found that free streaming makes collapse harder [1]. The threshold on the linear compaction C_l, which sets abundance, rises by $\Delta C_{l,c}/C_{l,c} \simeq 0.235\, R_\nu$, and the threshold δ_c on the compaction peak rises by $\simeq 0.15\, R_\nu$. For Gaussian fluctuations a perfect-fluid collapse probability β becomes about $\beta^{1.18}$ above $6 \times 10^5\, M_\odot$.

He also ran gradual decoupling, with collision rate $\kappa = H(\epsilon_{\rm dec}/\epsilon)^3$, where $\epsilon = 1/(aHr_m)$ is the Hubble radius in units of the perturbation size. Write G for the share of the instantaneous free-streaming shift that survives:

$$G = \frac{\delta_c - \delta_c^{\rm fluid}}{\delta_c^{\rm inst} - \delta_c^{\rm fluid}}. \tag{1}$$

His Table II gives $G = 0.959$, 0.867, 0.683, 0.431, 0.190 and 0.063 at $\epsilon_{\rm dec} = 0.25$, 0.5, 1, 2, 4 and 8. He notes that linear theory gives estimates of opposite sign and "does not determine which diagnostic, if any, predicts the nonlinear threshold" [1, Sec. IV C].

Two questions follow and are answered here. What in the neutrino transport operator carries the effect (Section 2)? And which clock, and which part of the decoupling history, sets G (Sections 3–5)?

2 The collision operator as a cut

For one Fourier mode k, the linearised neutrino intensity in McCormick's relaxation model evolves under

$$\partial_t f = Lf, \qquad L = A + D, \qquad A = -ik\mu, \qquad D = -\kappa(I - P), \tag{2}$$

where μ is the cosine of the angle to the wave vector and P projects onto the isotropic part. Free streaming A is anti-Hermitian. Collisions D are Hermitian, conserve the isotropic part (the neutrino energy in the plasma frame) and remove the flux, which goes to the plasma. In the normalised Legendre basis $|\ell\rangle$, streaming couples only neighbours, $\langle \ell+1|\mu|\ell\rangle = c_\ell = (\ell+1)/\sqrt{(2\ell+1)(2\ell+3)}$, so the operator lives on a chain $\ell = 0, 1, 2, \ldots$ (Figure 1).

Non-normality is connection energy. For Hermitian D and anti-Hermitian A, $[L, L^*] = 2[A, D]$, and for diagonal D, $[A, D]_{ij} = A_{ij}(D_j - D_i)$. Hence

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \sum_{\text{links } \ell \to \ell+1} k^2 c_\ell^2\, (D_{\ell+1} - D_\ell)^2. \tag{3}$$

This is the scalar case of the identity in [3]: the self-commutator energy of a transport operator is the energy of its Hermitian field failing to be parallel along the links [derived; Appendix A.1]. With one rate, only the link $0 \to 1$ has a jump, so

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \frac{\kappa^2 k^2}{3} \qquad \text{[derived; checked to } 10^{-12}\text{, I1]}. \tag{4}$$

All the non-normality of the single-rate model lies on the link between density and flux, the same channel as the momentum drag κq in McCormick's plasma equation [1, Eq. (6)].

Beyond relaxation time. Let the flux relax at κ_1 (momentum transfer) and shear and higher moments at κ_2. Then two links carry jumps:

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \frac{k^2\kappa_1^2}{3} + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2 \qquad \text{[derived; checked, I2]}. \tag{5}$$

The energy is stationary in κ_2 at $\kappa_2 = \kappa_1$. If neutrino–neutrino scattering damps shear but not flux, with rates relative to $e^\pm$ scattering of 1.5 for ν_e and 7 for $\nu_{\mu,\tau}$, the shear link carries 64% and 97.5% of the non-normality [computed, under those assumed ratios].

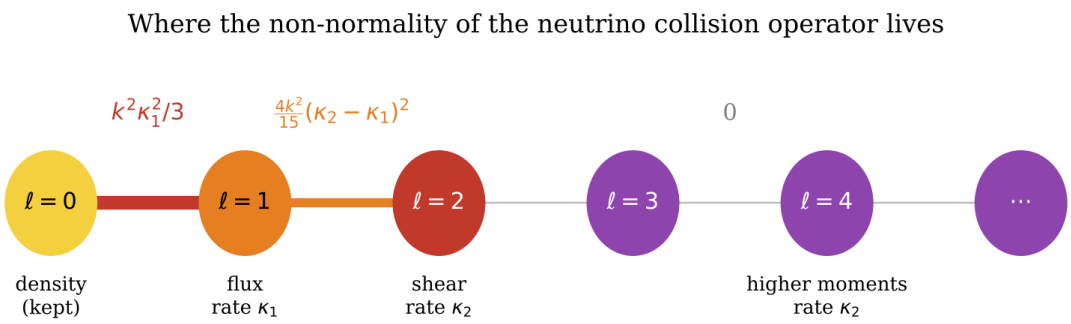


Figure 1: The neutrino moments form a chain. Streaming links each moment to its neighbours; collisions put a rate on each. Non-normality sits only where neighbouring rates differ: on the density–flux link ($k^2\kappa_1^2/3$) and, if shear relaxes at its own rate, on the flux–shear link. Every other link carries none.

The weight across the cut is the sound speed. With $H = k\mu$ the weight between kept and streaming parts is $E_B = \|PH(I-P)\|^2 = k^2c_0^2 = k^2/3$, so $\sqrt{E_B} = k/\sqrt{3} = c_s k$, the radiation sound speed [4] [derived; I3]. A rate bound built on E_B therefore sees the root-mean-square streaming speed $1/\sqrt{3}$, not the light-speed front that empties the overdensity [5].

The kept sector, exactly. The isotropic part has the closed propagator

$$\langle 0|(s-L)^{-1}|0\rangle = \frac{g}{1-\kappa g}, \qquad g = \frac{1}{k}\arctan\frac{k}{s+\kappa} \qquad \text{[derived; checked, I4].} \tag{6}$$

The two-moment truncation, keeping $\ell = 0, 1$, has eigenvalues $-\kappa/2 \pm \sqrt{\kappa^2/4 - k^2/3}$, which merge at $\kappa = 2k/\sqrt{3}$ [derived; I5]. This exceptional point belongs to the truncation. With 200 moments the least-damped mode turns real only at $\kappa \approx 0.646\,k$, and in relaxation kinetic theory the hydrodynamic poles meet a branch cut instead [8, 9] [computed].

3 New runs of McCormick's code

The code [2] was run unchanged except for one copied file in which the decoupling exponent n in $\kappa = H(\epsilon_{\rm dec}/\epsilon)^n$ is an input. Pure radiation, $R_\nu = 0.409$, $N = 400$ cells, $N_\mu = 32$ angle bins, $\epsilon_0 = 0.1$, as in his Table II. A reproduction run at $\delta = 0.5109$ with $n = 3$, $\epsilon_{\rm dec} = 2$ forms a black hole, inside his published bracket $(0.51074, 0.51109)$ [measured; Appendix B.4]. For the steep history, κ/H was capped at 5×10^5, the stiffness his own $\epsilon_{\rm dec} = 8$ runs reach at ϵ_0. At that cap deviations from isotropy are of relative order $H/\kappa \le 2 \times 10^{-6}$, the same estimate McCormick uses for his switch to transport [1, Sec. III].

For every test below each prediction and its kill condition were written before the runs (Appendix B.3). The threshold search runs two points per round; a run that neither collapses nor disperses by the end time, or that stops on an error, never moves the bracket, is retried once with a longer end time and is reported if still undecided. No point in any run below was undecided.

Test A: a profile that collapses late. The broad profile $\alpha = 0.3$ is decided near $\epsilon \approx 7$–9, against $\epsilon \approx 4$ for the Gaussian. Its thresholds are $\delta_c^{\rm fluid} = 0.44805$ and $\delta_c^{\rm inst} = 0.47747$ [2]. With $n = 3$:

	δ_c	G (broad)	G (Gaussian, [1])
$\epsilon_{\rm dec} = 4$	0.45337 ± 0.00013	0.181 ± 0.004	0.190
$\epsilon_{\rm dec} = 8$	0.44938 ± 0.00012	0.045 ± 0.004	0.063

The broad profile keeps the same share of the neutrino effect as the Gaussian or less (29% less at $\epsilon_{\rm dec} = 8$), although it collapses about twice as late (Figure 2) [measured]. A clock tied to the collapse time would have raised G; it fell. Because the test changes the profile shape together with the collapse time, it shows that the collapse time alone does not set the clock; it does not by itself separate the two, and it does not fix which earlier epoch does.

Tests B, C and D: a steeper decoupling history. Gaussian profile, $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^6$:

	δ_c	G ($n = 6$)	G ($n = 3$, [1])
C: $\epsilon_{\rm dec} = 1$	0.51756 ± 0.00019	0.648 ± 0.006	0.683
B: $\epsilon_{\rm dec} = 2$	0.50743 ± 0.00011	0.317 ± 0.004	0.431
D: $\epsilon_{\rm dec} = 3$	0.50245 ± 0.00019	0.154 ± 0.006	≈ 0.30

[measured]. The $n = 3$ value at $\epsilon_{\rm dec} = 3$ lies between McCormick's 0.431 and 0.190 at 2 and 4. A neutrino population that decouples more abruptly, but by the same nominal time, keeps less of its effect, and the gap widens with $\epsilon_{\rm dec}$. The history matters, not only its end point.

Numerical controls. Test B was repeated with the rate cap raised tenfold, to 5×10^6, and on a finer grid, $N = 600$ cells and $N_\mu = 48$ angle bins. The thresholds are $\delta_c = 0.50740 \pm 0.00011$ and 0.50751 ± 0.00006, shifts of -0.00003 and $+0.00008$ against the pre-registered tolerance of 0.0005 [measured]. The cap acts only while the neutrinos are tightly coupled: the uncapped rate starts at $(\epsilon_{\rm dec}/\epsilon_0)^6$ and falls below the cap at $\epsilon \approx 0.11$, 0.22 and 0.34 for $\epsilon_{\rm dec} = 1$, 2 and 3. The cap control was run at $\epsilon_{\rm dec} = 2$ only; at $\epsilon_{\rm dec} = 1$ and 3 it is untested.

4 What does not set the threshold

Each reading below was tested against numbers. Only the last two were tested against runs set up for them in advance; the first three were tested against McCormick's published curve.

1. **The first maximum of linear theory.** The linear density contrast at its first maximum ($\epsilon \approx 1.7$) reproduces the instantaneous shift (centre factor 0.944, as McCormick finds). Under gradual decoupling the share it keeps is 0.86, 0.59, 0.19, -0.05, -0.02 and 0.00, a largest error of 0.50. Decoupling after $\epsilon \approx 1.7$ is nearly invisible to it, yet still moves the nonlinear threshold [computed].

2. **A sharp wall, and rate ratios.** Treat the two-moment exceptional point, crossed at $\epsilon_{\rm EP} = 0.811\,\epsilon_{\rm dec}^{3/4}$ for $kr_m = 2$, as a wall, or use the parameter-free escape fractions $k/(k+\kappa)$, $e^{-\kappa/k}$ and $1 - e^{-k/\kappa}$. Averaging each over two windows, with two weightings and two scales, gives 32 combinations. The wall variants miss by 0.14 or more. With the window starting at horizon entry, every combination misses by 0.17 or more. One combination, $e^{-\kappa/k}$ averaged

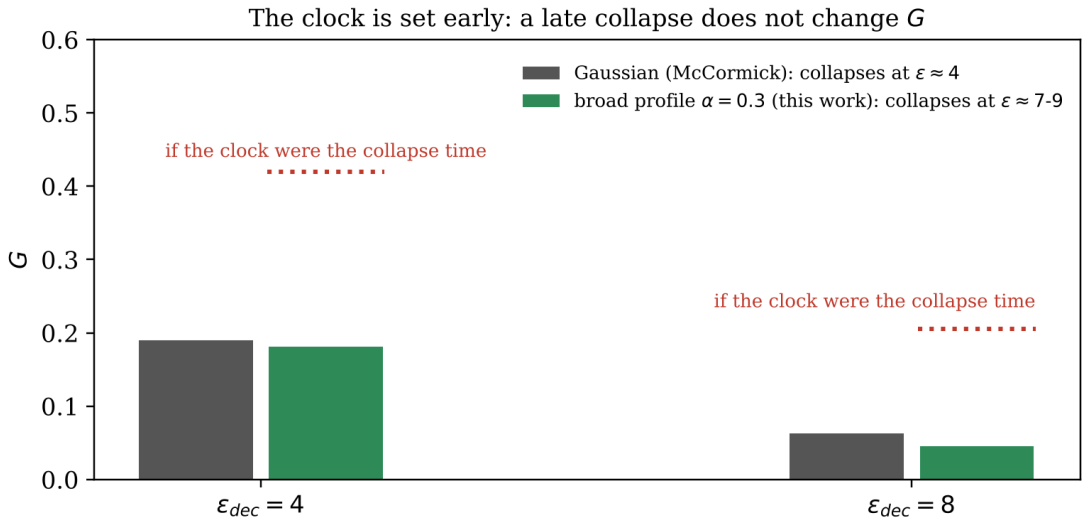


Figure 2: Test A. The broad profile collapses at $\epsilon \approx 7$–9, yet keeps the same share G of the free-streaming shift as the Gaussian. A clock tied to the collapse time would have given the dotted values.

from ϵ_0, comes within 0.077; selected from 32, it is not counted. On Test B, which it never saw, it predicts 0.513 against 0.317 [computed; measured].

3. **Diffusive escape.** Diffusion with $D = 1/(3\sigma)$ from the conformal rate $\sigma = \epsilon_{\rm dec}^3/\tau^4$, combined in series with streaming at the collapse time ϵ^*, gives $G = 1/[1+\sqrt{15}(\epsilon_{\rm dec}/\epsilon^*)^{3/2}]$. With $\epsilon^* = 4$ it matches McCormick's curve to 0.02, but it was built after seeing that curve, and a free fit of its exponent gives 1.61 ± 0.03 rather than 3/2. Test A with $\epsilon^* = 7$–9 predicted 0.37–0.47 and 0.17–0.24, and Test B predicted 0.62. Both are excluded [measured].

4. **History-blind.** If only $\epsilon_{\rm dec}$ mattered, Tests B and D would give 0.431 and about 0.30. They give 0.317 and 0.154; Test D misses by 0.146 against a kill line of 0.08 [measured].

5 What survives: the memory of the kept sector

The reading that survives treats the isotropic part as a kept sector and asks how much of it the decoupling operator keeps [4]. Start a mode in $|0\rangle$ at $\tau_0 = 0.02$ and evolve it under $\partial_\tau v = -(ik\mu + a\kappa(\tau)(I-P))v$, with $a\kappa = \epsilon_{\rm dec}^n/\tau^{n+1}$. Let $A(\tau)$ be the population left in $|0\rangle$, averaged over the Gaussian profile with weights $k^2 e^{-k^2/2}$. Define

$$g(\epsilon^*) = \frac{1 - A_{\epsilon_{\rm dec}}(\epsilon^*)}{1 - A_{\rm inst}(\epsilon^*)}, \tag{7}$$

the share of the kept sector's loss that has happened by the readout time ϵ^*.

With no fitted number, read at the first compaction maximum $\epsilon^* = 1.7$, it gives 0.941, 0.860, 0.671, 0.344, 0.087 and 0.013 against McCormick's G. That is within 0.02 for $\epsilon_{\rm dec} \le 1$, with a largest error of 0.103, just outside the 0.10 bar set for it in advance [computed]. McCormick's own curve passes $G = \frac{1}{2}$ at $\epsilon_{\rm dec} \approx 1.67$, within 3% of the first maximum: the neutrinos must be decoupled by then for half the effect to survive.

The late shortfall says the collapse is still listening after the first maximum. Mixing the two epochs McCormick's numbers single out, $g_2 = w\,g(1.7) + (1-w)\,g(4.0)$, with one weight fitted to his curve, $w = 0.89$, gives a largest error of 0.022 [fit]. The form is not unique: with the scale fixed at $kr_m = 2$ the single-epoch reading misses by 0.093, and the best single readout epoch, $\epsilon^* = 1.87$, by 0.065. Its weight does not carry over between weightings: the same w with the scale fixed at $kr_m = 2$ misses by 0.115, outside the 0.10 bar declared for that check [computed]. The form is a fitted shape with a stated sensitivity, not a derived law.

The test is what that fitted form says about histories it never saw. All its $n = 6$ values were computed in one file dated 13:47 on 30 October, before any $n = 6$ run, and were not refitted afterwards:

	predicted	measured	miss	history-blind	miss
C: $\epsilon_{\rm dec} = 1$	0.663	0.648	0.014	0.683	0.035
B: $\epsilon_{\rm dec} = 2$	0.246	0.317	0.071	0.431	0.114
D: $\epsilon_{\rm dec} = 3$	0.118	0.154	0.037	≈ 0.30	0.146

Every miss of the memory form lies inside the kill line of 0.08 set before its run; the history-blind rule fails at $\epsilon_{\rm dec} = 3$ (Figures 3 and 4) [measured]. At $\epsilon_{\rm dec} = 2$ and 3 the form falls below the measurement: it counts slightly too much of the early erasure. Test C was set where the two rules nearly agree, as a survival check; Test D was set where they separate. A readout window set by the neutrino flux and the enclosed mass between first maximum and apparent horizon, as McCormick suggests [1, Sec. IV C], is the natural next object. It is untested.

6 Where the memory sits

A fluid closure carries the density, the flux and the anisotropic stress, the moments $\ell = 0, 1, 2$. The memory form of Section 5 says the threshold depends on how the kept sector was emptied. The question here is whether that history can sit in moments a closure does not carry.

Exact elimination of the hidden moments. Split $v = (a, h)$ with $a = v_0$ and $h = (v_1, v_2, \dots)$. In the relaxation model every hidden moment is damped at the same rate, so the collision term on the hidden block is $\sigma(\tau)$ times the identity and commutes with streaming. The hidden propagator is therefore exact in closed form,

$$U_Q(t,r) = e^{-[\Sigma(t)-\Sigma(r)]}\, e^{-ik(t-r)M_{QQ}}, \qquad \Sigma(\tau) = -\frac{\epsilon_{\rm dec}^n}{n\tau^n}, \quad \Sigma' = \sigma, \tag{8}$$

and eliminating h gives one equation for the kept amplitude,

$$\dot a(t) = S(t) - k^2 \int_s^t K(t,r)\, a(r)\, dr, \qquad K = c_0^2\, [U_Q(t,r)]_{11}, \qquad S = -ikc_0\, [U_Q(t,s)h(s)]_1 \tag{9}$$

[derived; Appendix A.5]. The kernel K is the memory; the source S is everything the hidden moments held at time s.

The matched pair. Take two states with the same $(v_0, v_1, v_2) = (1, 0.2, 0.1)$, differing only by ± 0.05 in v_3 (or v_4), scaled by one constant to unit norm. Both are positive angular distributions ($f(\mu) \geq 0.50$). Their kept amplitudes start equal, so their difference obeys the equation above driven only by S of the hidden difference. Both are evolved under the same history, $\epsilon_{\rm dec} = 0.5$,

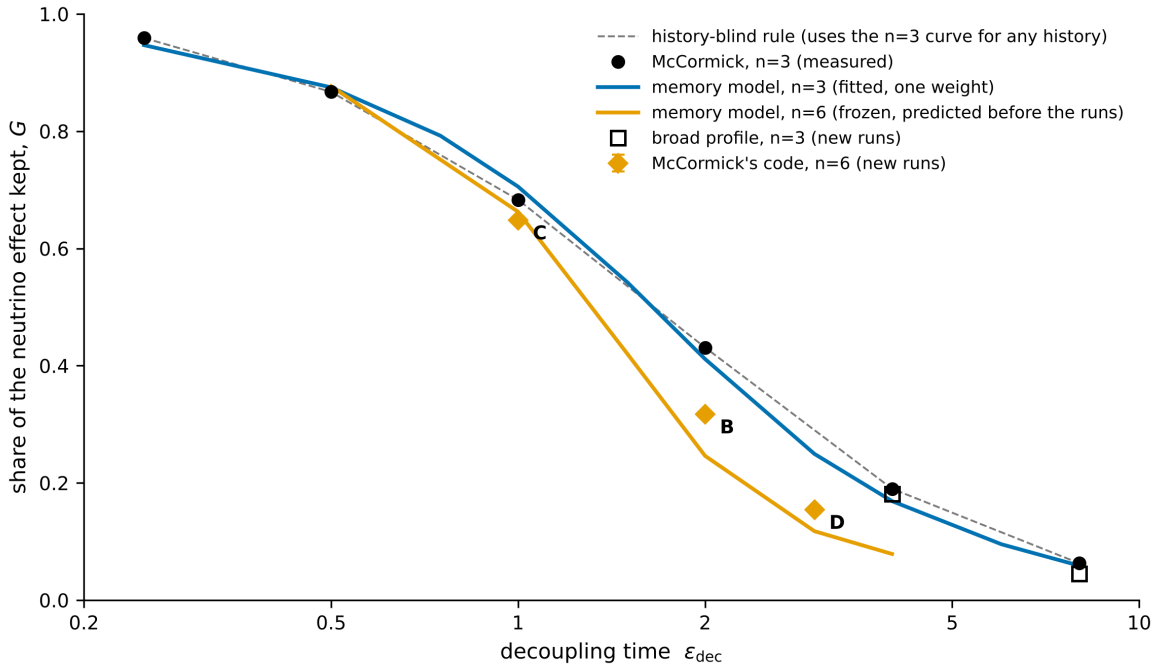


Figure 3: The surviving share G of the free-streaming shift against decoupling time. Black dots: McCormick's Gaussian runs with $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^3$; the grey dashed line through them is what a history-blind rule gives for any history. Blue: the kept-sector memory with one weight fitted to the black dots. Orange line: the same form for $\kappa/H = (\epsilon_{\rm dec}/\epsilon)^6$, computed before any such run and not refitted. Orange diamonds: the new $n = 6$ runs C, B and D. Open squares: the broad profile (Test A).

with the planting time τ_s scanned across decoupling. The readout is the profile-weighted change in (v_0, v_1, v_2), relative to its size, $R_{\rm ret}$; the witness is on if $R_{\rm ret} > 10^{-3}$. Every on/off call was written down before the run, from the survival factor of the hidden block, $\exp[-\epsilon_{\rm dec}^n/(n\tau_s^n)]$, and the leakage $(k/\sigma)^\ell$ through damped rungs (Appendix B.14).

$\tau_s/\epsilon_{\rm dec}$	$n=3,\ \ell=3$	$n=3,\ \ell=4$	$n=6,\ \ell=3$	$n=6,\ \ell=4$
0.1	7×10^{-15}	5×10^{-19}	8×10^{-24}	5×10^{-31}
0.25	6×10^{-10}	6×10^{-11}	2×10^{-15}	7×10^{-20}
0.5	$\mathbf{8.6\times10^{-3}}$	$\mathbf{7.9\times10^{-3}}$	3×10^{-6}	3×10^{-6}
1	**0.081**	**0.074**	**0.094**	**0.086**
2	**0.095**	**0.085**	**0.098**	**0.088**

$R_{\rm ret}$ at $\tau = 4$; bold entries are on [computed]. Every call came out as predicted. The survival law is quantitative: for $n = 3$ the ratio of the entries at 0.5 and 1 is 0.106 against $e^{-8/3}/e^{-1/3} = 0.097$, and for $n = 6$ at 0.5 the law gives 1.4×10^{-6} against 1.6×10^{-6} at $\tau = 1.7$. Planted in the retained moments instead, deep in coupling the order is $\ell = 1 > 2 > 3 > 4$, each rung costing about $10^{-4} \approx k/\sigma$; after decoupling all four are of the same size. A hidden difference set at decoupling changes the later density, flux and stress as much as a visible one (Figure 5).

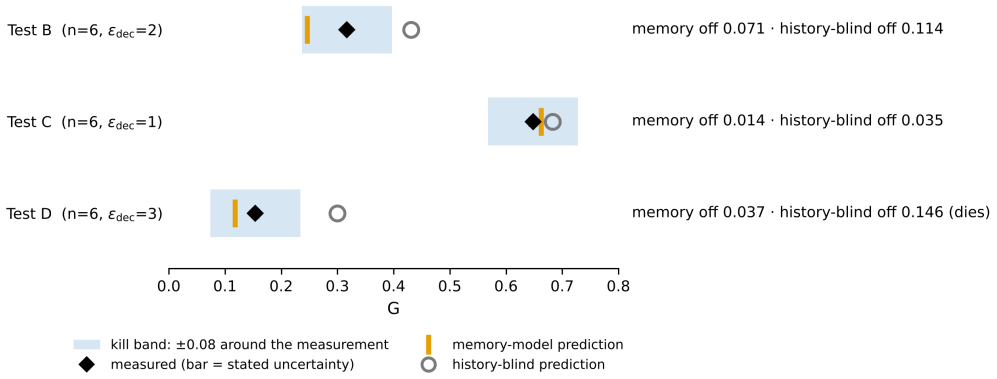


Figure 4: The three steep-history tests against their kill lines. The shaded band is ±0.08 around each measurement. Orange: the memory form's prediction, made before the run. Grey circle: the history-blind rule. At $\epsilon_{\rm dec} = 3$ the history-blind rule falls outside the band.

Checks. Doubling the number of moments, $60 \to 120$, and tightening the solver tolerances change every entry by at most 1.4×10^{-5} of its value. Solving the elimination equation from the closed-form U_Q alone, with no solve of the full system, reproduces the full difference to 3×10^{-7}–7×10^{-7}, falling fourfold with each halving of the step; leaving out the kernel misses by 19–234% [computed; Appendix B.14].

A speed limit on the kept population. Collisions conserve $|0\rangle$ and remove the rest, so $\|v\|$ never grows, and streaming links $|0\rangle$ only to $|1\rangle$, with weight $kc_0 = \sqrt{E_B}$. Hence for $A = |v_0|^2$

$$\dot{A}^2 \le 4E_B\, A(1-A), \qquad E_B = k^2/3 \tag{10}$$

[derived; Appendix A.6]. On every trajectory computed here the largest value of the ratio is 0.87 [computed]. The bound limits how fast the kept sector can be emptied; it does not set the threshold.

Reading. A closure on density, flux and stress is exact only for structure set while the neutrinos are tightly coupled. Structure set at or after decoupling survives in the finer angular moments and streams back into the visible ones by several per cent. The memory McCormick's runs record therefore has a place to live that no fluid closure contains. Whether realistic initial data carry such structure, and how far it moves the nonlinear threshold, are untested: the memory curves of Section 5 start with the hidden moments empty.

7 Tests for McCormick's code

1. **The annihilation control.** During $e^{\pm}$ annihilation the plasma sound speed falls to $c_s = 0.519$ at $T = 0.131\,\mathrm{MeV}$, while the neutrinos' root-mean-square speed stays $1/\sqrt{3}$. The contest between front and sound, $f = (1-c_s)/(1-1/\sqrt{3})$, reaches 1.135 at $1.6 \times 10^7\, M_\odot$ ($c_s = 0.521$ there). If the softer plasma enhanced the free-streaming effect, the shift would reach 7.5%, outside McCormick's envelope of 5.3–6.2%, so that reading already fails. If it reduces it, the shift stays at 5.5–6.3% [computed; reading]. At $1.6 \times 10^7\, M_\odot$ the two readings differ from the unmodified 6.31% by +13.5% and −11.9%. The run McCormick lists as missing, the same

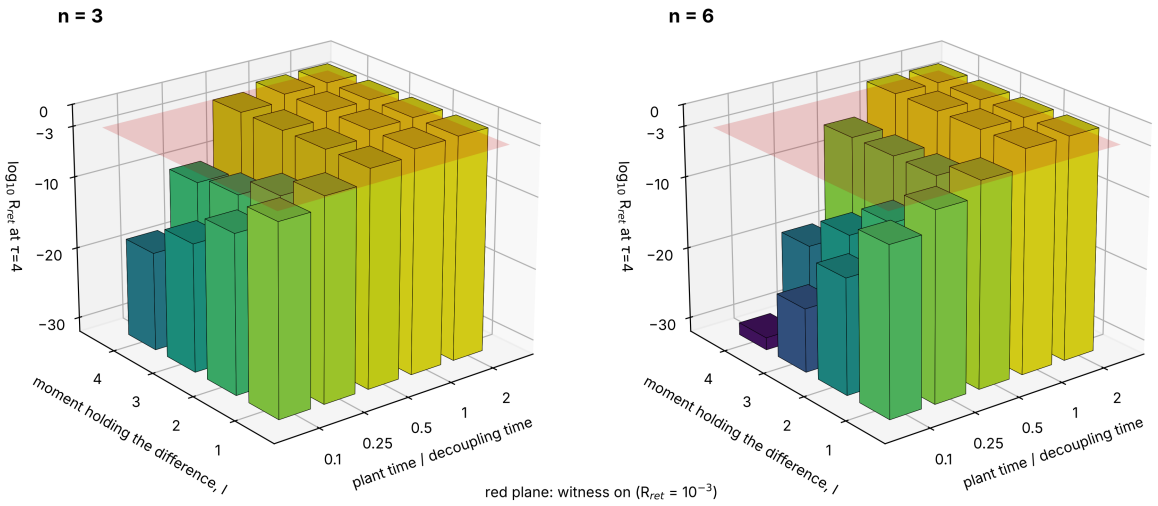


Figure 5: The door. Strength of the matched-pair witness, $\log_{10} R_{\mathrm{ret}}$ at $\tau = 4$, against the time the hidden difference was planted (in units of ϵ_{dec}) and the moment that held it. Planted deep in the coupled era the difference is erased, down to 10^{-30}; planted at or after decoupling it returns at the 10^{-1} level. The red plane is the witness threshold 10^{-3}. The steeper history ($n = 6$) closes the door more sharply.

transport at fixed R_ν with $p = e/3$ against the full equation of state at 3×10^6–$1.6 \times 10^7\, M_\odot$, would confirm the sign.

2. **Shear against flux.** By Section 2, a collision term that damps shear faster than flux puts most of the non-normality on the flux–shear link. Whether the threshold follows the shear rate more than the momentum-transfer rate needs the response itself, not the energy: the energy is stationary at $\kappa_2 = \kappa_1$, but the response need not be. With three moments and $k = s = \kappa_1 = \kappa_2 = 1$, the kept response is $R_{00} = 32/37$ with $\partial R_{00}/\partial\kappa_2 = -5/1369 \neq 0$, while $\partial E/\partial\kappa_2 = 0$ [derived; Appendix A.7]. The proposed run varies κ_2 at fixed κ_1 in McCormick's code and reads the threshold [proposed].

3. **Independence of R_ν.** The kept-sector form contains no R_ν. One run at $R_\nu = 0.2$, $\epsilon_{\mathrm{dec}} = 2$, $n = 3$ should give the form's own value, $G = 0.411$, which McCormick's $R_\nu = 0.409$ run puts at 0.431 [proposed].

4. **Which mean of the rates.** McCormick calibrates his single rate to an energy-group calculation. The arithmetic mean of the group rates reproduces his calibrated decoupling temperatures: 1.62 MeV against 1.6 ($e^{\pm}$ only), 1.02 MeV against 0.93–1.0 (with $\nu\nu$), and a mass step of 2.53 against 2.5–3. The harmonic and geometric means fail; the mean of squares also passes, so the class of means is identified, not one mean [4] [computed; reading].

8 Relation to other work

The closest published analogue is bulk viscosity, which raises the threshold roughly in proportion to its strength in full numerical relativity [10]. Free streaming is a second dissipative channel with the same sign. Papanikolaou fixes thresholds at horizon crossing for a time-dependent equation of

state, with δ_c following c_s^2 [11]. Escrivà ties them to the initial profile and the curvature of the compaction at its peak [12]. Test A is consistent with both: the neutrino clock is set early.

The 2026 thermal-history mass functions through neutrino decoupling and $e^{\pm}$ annihilation treat neutrinos as part of the fluid [13], so the step McCormick found is absent from them. For supermassive seeds, the μ-distortion bound leaves only heavy-tailed statistics [14], for which McCormick finds that the raised threshold costs a factor 1.2–8 in abundance in his examples [1].

A search of the literature to October 2026 found no other work on thresholds with free-streaming components, on gradual kinetic decoupling in numerical relativity, or on how the threshold depends on when a collision rate switches off. Tests A–D are, as far as that search shows, the first such runs.

9 Summary

The neutrino collision operator is a cut whose whole non-normality sits on one link, density to flux, with weight equal to the sound speed. McCormick's code, run on five cases outside his paper, shows that the collapse time alone does not set the neutrino clock, and that the decoupling history changes the threshold, not only its end point. The kept-sector memory, read mostly at the first compaction maximum and fitted once, predicted three histories it had not seen to within 0.071, 0.014 and 0.037, while a history-blind rule failed. The steep-history thresholds do not move with the rate cap or the grid. The memory sits in the angular structure a fluid closure drops: set at or after decoupling, it returns into density, flux and stress; set before, it is erased.

References

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A Proofs

A.1 Non-normality is connection energy, for the neutrino chain

Let $L = A + D$ with $A^* = -A$ and $D^* = D$. Then $L^* = D - A$, and

$$[L, L^*] = (A + D)(D - A) - (D - A)(A + D) = 2(AD - DA) = 2[A, D].$$

For diagonal $D = \mathrm{diag}(D_\ell)$, $[A, D]_{ij} = A_{ij}(D_j - D_i)$. So

$$\tfrac{1}{8}\|[L, L^*]\|_F^2 = \tfrac{1}{2}\sum_{i\neq j} |A_{ij}|^2 (D_i - D_j)^2.$$

Streaming couples only ℓ and $\ell \pm 1$, with $|A_{\ell,\ell+1}| = kc_\ell$. Each link appears twice in the ordered sum, which gives $\sum_\ell k^2 c_\ell^2 (D_{\ell+1} - D_\ell)^2$. With $D = (0, -\kappa_1, -\kappa_2, -\kappa_2, \dots)$, $c_0^2 = 1/3$ and $c_1^2 = 4/15$, this is $k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$. With $\kappa_1 = \kappa_2 = \kappa$ it is $\kappa^2 k^2/3$. □

A.2 The weight across the cut

$PH(I - P)$ with $H = k\mu$ has the single entry $\langle 0|k\mu|1\rangle = kc_0$, so $E_B = k^2/3$. The variance of $k\mu$ in the isotropic state is $k^2\langle\mu^2\rangle = k^2/3$, the same number. □

A.3 The kept-sector propagator

Write $u = (s - L)^{-1}|0\rangle$, so $(s+\kappa+ik\mu)u = |0\rangle + \kappa Pu$. Projecting onto $|0\rangle$ with $\langle 0|(s+\kappa+ik\mu)^{-1}|0\rangle = \frac{1}{2}\int_{-1}^{1} d\mu/(s + \kappa + ik\mu) = k^{-1}\arctan[k/(s + \kappa)] = g$ gives $u_0 = g(1 + \kappa u_0)$, so $u_0 = g/(1 - \kappa g)$. □

A.4 The truncated exceptional point

On $\mathrm{span}\{|0\rangle, |1\rangle\}$, $L = \begin{pmatrix} 0 & -ik/\sqrt{3} \\ -ik/\sqrt{3} & -\kappa \end{pmatrix}$, with characteristic polynomial $\lambda^2 + \kappa\lambda + k^2/3$. Its discriminant $\kappa^2 - 4k^2/3$ vanishes at $\kappa = 2k/\sqrt{3}$, where the matrix is not diagonalisable. □

A.5 Exact elimination of the hidden moments

Write $\partial_\tau v = -(ikM + \sigma(\tau)Q)v$ with $Q = I - P$, and split $v = (a, h)$ along P and Q. Then $\dot{a} = -ikc_0 h_1$ and $\dot{h} = -(ikM_{QQ} + \sigma)h - ikc_0\, a\, e_1$. On the hidden block $\sigma(\tau)I$ commutes with M_{QQ}, so the generators at different times commute and the propagator is the product $U_Q(t, r) = e^{-\int_r^t \sigma}\, e^{-ik(t-r)M_{QQ}}$, with $\int_r^t \sigma = \Sigma(t) - \Sigma(r)$, $\Sigma(\tau) = -\epsilon_{\mathrm{dec}}^n/(n\tau^n)$. Variation of constants gives $h(t) = U_Q(t, s)h(s) - ikc_0 \int_s^t U_Q(t, r)e_1 a(r)\, dr$; substituting into $\dot{a}$ gives

$$\dot{a}(t) = -ikc_0[U_Q(t, s)h(s)]_1 - k^2c_0^2 \int_s^t [U_Q(t, r)]_{11} a(r)\, dr.$$

For two states with equal $a(s)$ the difference Δa obeys the same equation with source $S[\Delta h(s)]$ and $\Delta a(s) = 0$; if $\Delta h(s) = 0$ it vanishes identically. The time-independent case $e^{t(X-\sigma I)} = e^{-\sigma t}e^{tX}$ is stated in Lean (Appendix C). □

A.6 The speed limit on the kept population

With $D = -\sigma Q \le 0$, $\frac{d}{d\tau}\|v\|^2 = -2\sigma\|Qv\|^2 \le 0$, so $\|v\| \le 1$ for a unit initial state. Since $\langle 0|D = 0$ and $\langle 0|M = c_0\langle 1|$, $\dot{v}_0 = -ikc_0 v_1$, and

$$\dot{A} = 2\,\mathrm{Re}\big(\bar{v}_0\, \dot{v}_0\big) = 2\,\mathrm{Re}\big(\bar{v}_0(-ikc_0)v_1\big), \qquad |\dot{A}| \le 2kc_0|v_0||v_1|.$$

With $|v_1|^2 \le \|v\|^2 - |v_0|^2 \le 1 - A$ and $k^2c_0^2 = E_B$, $\dot{A}^2 \le 4E_B A(1 - A)$. The core inequality $|2\,\mathrm{Re}(\bar{a}(-ih)b)| \le 2|h||a||b|$ and the corollary under $|a|^2 + |b|^2 \le 1$ are stated in Lean (Appendix C). □

A.7 Energy stationarity is not response stationarity

Keep three moments, $\ell = 0, 1, 2$, with streaming weights $c_0^2 = 1/3$, $c_1^2 = 4/15$ and rates $(0, \kappa_1, \kappa_2)$. The kept response $R_{00} = \langle 0|(s + ikM + D)^{-1}|0\rangle$ is the continued fraction

$$R_{00} = \cfrac{1}{s + \cfrac{k^2c_0^2}{s + \kappa_1 + \cfrac{k^2c_1^2}{s + \kappa_2}}}.$$

At $k = s = \kappa_1 = 1$ this is $R_{00}(\kappa_2) = \big(1 + \frac{1}{3}/(2 + \frac{4/15}{1+\kappa_2})\big)^{-1}$, so $R_{00}(1) = 32/37$ and $R_{00}'(1) = -5/1369$. The energy $E = k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$ has $\partial E/\partial\kappa_2 = \frac{8k^2}{15}(\kappa_2 - \kappa_1) = 0$ there. Checked by direct inversion of the 3×3 matrix; stated in Lean (Appendix C). □

B Evidence: scripts, run logs and outputs

B.1 Identity checks

```
# SCRIPT: PBH-NU-IDENTITIES  checks of every derived identity used in the paper
# Kill conditions (written first): each identity must hold to 1e-10 relative, else the statement is withdrawn.
import numpy as np
from scipy.integrate import quad
LM=80
c=lambda l:(l+1)/np.sqrt((2*l+1)*(2*l+3))          # <l+1|mu|l> in the normalised Legendre basis
Mu=np.zeros((LM+1,LM+1))
```

```
for l in range(LM): Mu[l,l+1]=Mu[l+1,l]=c(l)
def L_op(k,k1,k2):
    D=np.diag([0.0,-k1]+[-k2]*(LM-1)); A=-1j*k*Mu; return A,D,A+D
fro2=lambda X:np.sum(np.abs(X)**2)
print('I1  single rate: (1/8)|[L,L*]|^2 = kappa^2 k^2/3')
for k,kap in [(1.0,0.3),(2.0,1.5),(0.7,4.0)]:
    A,D,L=L_op(k,kap,kap); lhs=fro2(L@L.conj().T-L.conj().T@L)/8
    print('    k=%.1f kappa=%.1f  lhs=%.12f  rhs=%.12f'%(k,kap,lhs,kap**2*k**2/3))
print('I2  split rates: (1/8)|[L,L*]|^2 = k^2 k1^2/3 + (4k^2/15)(k2-k1)^2')
for k,k1,k2 in [(1.0,0.3,2.0),(2.0,1.5,0.4),(0.7,4.0,9.0)]:
    A,D,L=L_op(k,k1,k2); lhs=fro2(L@L.conj().T-L.conj().T@L)/8
    print('    k=%.1f k1=%.1f k2=%.1f lhs=%.12f rhs=%.12f'%(k,k1,k2,lhs,k**2*k1**2/3+4*k**2/15*(k2-k1)**2))
print('I3  weight across the cut: |P (k mu) (1-P)|^2 = k^2/3, so sqrt(E_B) = k/sqrt3 = c_s k')
P=np.zeros((LM+1,LM+1)); P[0,0]=1; Q=np.eye(LM+1)-P
for k in [1.0,2.5]: print('    k=%.1f  E_B=%.12f  k^2/3=%.12f'%(k,fro2(P@(k*Mu)@Q),k**2/3))
print('I4  monopole propagator <0|(s+i k mu+kappa(1-P))^-1|0> = g/(1-kappa g), g=arctan(k/(s+kappa))/k')
for k,kap,s in [(1.0,0.5,0.3),(3.0,2.0,0.1),(0.5,0.2,1.0)]:
    Lfull=s*np.eye(LM+1)+1j*k*Mu+kap*Q; num=np.linalg.solve(Lfull,np.eye(LM+1)[:,0])[0]
    g=np.arctan(k/(s+kap))/k; print('    k=%.1f kappa=%.1f s=%.1f  matrix=%.12f  closed=%.12f'%(k,kap,s,num.
     real,(g/(1-kap*g))))
print('I5  two-moment truncation: eigenvalues of [[0,-ik/sqrt3],[-ik/sqrt3,-kappa]] merge at kappa=2k/sqrt3')
k=1.0; kap=2*k/np.sqrt(3); ev=np.linalg.eigvals(np.array([[0,-1j*k/np.sqrt(3)],[-1j*k/np.sqrt(3),-kap]]));
     print('    eigenvalues',np.round(ev,10))
```

```
I1  single rate: (1/8)|[L,L*]|^2 = kappa^2 k^2/3
    k=1.0 kappa=0.3  lhs=0.030000000000  rhs=0.030000000000
    k=2.0 kappa=1.5  lhs=3.000000000000  rhs=3.000000000000
    k=0.7 kappa=4.0  lhs=2.613333333333  rhs=2.613333333333
I2  split rates: (1/8)|[L,L*]|^2 = k^2 k1^2/3 + (4k^2/15)(k2-k1)^2
    k=1.0 k1=0.3 k2=2.0 lhs=0.800666666667 rhs=0.800666666667
    k=2.0 k1=1.5 k2=0.4 lhs=4.290666666667 rhs=4.290666666667
    k=0.7 k1=4.0 k2=9.0 lhs=5.880000000000 rhs=5.880000000000
I3  weight across the cut: |P (k mu) (1-P)|^2 = k^2/3, so sqrt(E_B) = k/sqrt3 = c_s k
    k=1.0  E_B=0.333333333333  k^2/3=0.333333333333
    k=2.5  E_B=2.083333333333  k^2/3=2.083333333333
I4  monopole propagator <0|(s+i k mu+kappa(1-P))^-1|0> = g/(1-kappa g), g=arctan(k/(s+kappa))/k
    k=1.0 kappa=0.5 s=0.3  matrix=1.623370180076  closed=1.623370180076
    k=3.0 kappa=2.0 s=0.1  matrix=0.889069885739  closed=0.889069885739
    k=0.5 kappa=0.2 s=1.0  matrix=0.937653083725  closed=0.937653083725
I5  two-moment truncation: eigenvalues of [[0,-ik/sqrt3],[-ik/sqrt3,-kappa]] merge at kappa=2k/sqrt3
    eigenvalues [-0.57735027+9.2e-09j -0.57735027-9.2e-09j]
```

B.2 The one change to McCormick's code

In a copy of `run_nuc.py`, the collision rate line became
`kap = (1/(2*t))*min((eps_dec/(2*np.sqrt(t*t0)))**nexp, kapcap)`
with `nexp` (default 3) and `kapcap` $= 5 \times 10^5$ as inputs. Nothing else was changed.

```
# SCRIPT: BEYOND-STEVE-BISECT  threshold bisection with gradual decoupling, profile alpha and rate exponent as
     inputs
import sys, json, time, os
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
from run_nuc_n import run
tag=sys.argv[1]; qshape=float(sys.argv[2]); epsd=float(sys.argv[3]); nexp=float(sys.argv[4])
lo,hi,tol=float(sys.argv[5]),float(sys.argv[6]),float(sys.argv[7]); emax=float(sys.argv[8])
log=[]; t0=time.time()
while hi-lo>tol:
    mid=0.5*(lo+hi); t1=time.time()
    out=run(mid,fnu=0.409,N=400,K=32,eps0=0.1,eps_dec=epsd,eps_max=emax,qshape=qshape,nexp=nexp)
    print(f'{tag} delta={mid:.5f}: {out["status"]} eps={out["eps"]:.2f} ({time.time()-t1:.0f}s)',flush=True)
    log.append((mid,out['status'],float(out['eps'])))
    if out['status']=='BH': hi=mid
    elif out['status']=='dispersed': lo=mid
    else: break
```

```
print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} ({(time.time()-t0)/60:.1f} min)',flush=
    True)
json.dump(dict(tag=tag,qshape=qshape,eps_dec=epsd,nexp=nexp,dc=0.5*(lo+hi),w=hi-lo,log=log),open(f'res_{tag}.
    json','w'))
```

```
# SCRIPT: BEYOND-STEVE-TRISECT  two trial amplitudes per round in parallel (thirds of the bracket)
import sys, json, time, os
from multiprocessing import Pool
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))
from run_nuc_n import run
def one(a):
    d,q,ed,n,em=a; o=run(d,fnu=0.409,N=400,K=32,eps0=0.1,eps_dec=ed,eps_max=em,qshape=q,nexp=n); return d,o['
     status'],float(o['eps'])
if __name__=='__main__':
    tag=sys.argv[1]; q=float(sys.argv[2]); ed=float(sys.argv[3]); n=float(sys.argv[4]); lo,hi,tol=float(sys.
     argv[5]),float(sys.argv[6]),float(sys.argv[7]); em=float(sys.argv[8])
    log=[]; t0=time.time()
    with Pool(2) as p:
        while hi-lo>tol:
            a=lo+(hi-lo)/3; b=lo+2*(hi-lo)/3
            res=p.map(one,[(a,q,ed,n,em),(b,q,ed,n,em)])
            for d,s,e in res: print(f'{tag} delta={d:.5f}: {s} eps={e:.2f}',flush=True); log.append((d,s,e))
            (da,sa,_),(db,sb,_)=res
            if 'BH' not in (sa,sb) and 'dispersed' not in (sa,sb): break
            if sa=='BH': hi=da
            elif sb=='BH': lo,hi=da,db
            else: lo=db
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} ({(time.time()-t0)/60:.1f} min)',
     flush=True)
    json.dump(dict(tag=tag,dc=0.5*(lo+hi),w=hi-lo,log=log),open(f'res_{tag}.json','w'))
```

B.3 Predictions written before the runs

```
BEYOND-STEVE pre-registration, written 2026-10-30 before any of these runs.
G = (delta_c - delta_c_fluid) / (delta_c_instant - delta_c_fluid), same profile.
Gaussian (alpha=1): fluid 0.49773, instant 0.52832 (Steve Table II).
alpha=0.3: fluid 0.44805, instant 0.47747 (Steve res_fl_q03 / res_nu409_q03).

TEST A  broad profile alpha=0.3, rate kappa=H(eps_dec/eps)^3, eps_dec = 4 and 8.
  Diffusive-escape law G=1/(1+sqrt15 (eps_dec/eps*)^1.5), eps* = 7-9 (alpha=0.3 decided late):
     G(4) = 0.38-0.46 ; G(8) = 0.17-0.24
  Null (curve does not move with eps*): G(4)=0.19, G(8)=0.063
  Kill: the law dies if G(4) < 0.30 or G(8) < 0.13.  Null dies if G(4) > 0.28.

TEST B  Gaussian, steeper rate kappa=H(eps_dec/eps)^6, eps_dec = 2.
  Diffusive-escape law with n=6: G = 1/(1+sqrt24 (eps_dec/eps*)^3), eps*=4  ->  G = 0.62
  Memory model (Weight Between agent): G = 0.25
  History-blind model: G = 0.43
  Kill: each model dies if measured G is more than 0.08 from its number.
```

B.4 Run logs

```
Reproduction check, 2026-10-30 ~14:25 CT, printed by the session (not logged to file at the time):
repro check n=3 eps_dec=2 at 0.5109 (Steve threshold 0.51092): BH 6.6146961105642985 90
(status BH, apparent horizon at eps 6.61, 90 s run time)
McCormick logG.txt bracket for the same case: 0.51109 BH, 0.51074 dispersed.
```

```
A_q03_ed4 delta=0.46300: BH eps=4.70 (116s)
A_q03_ed4 delta=0.45500: BH eps=6.24 (133s)
A_q03_ed4 delta=0.45100: dispersed eps=7.01 (145s)
A_q03_ed4 delta=0.45300: dispersed eps=7.97 (154s)
A_q03_ed4 delta=0.45400: BH eps=6.89 (141s)
```

```
A_q03_ed4 delta=0.45350: BH eps=7.63 (153s)
A_q03_ed4 delta=0.45325: dispersed eps=9.51 (176s)
RESULT A_q03_ed4: delta_c = 0.45337 +- 0.00013 (17.0 min)
```

```
A_q03_ed8 delta=0.46300: BH eps=4.41 (539s)
A_q03_ed8 delta=0.45500: BH eps=5.30 (545s)
```

```
A_q03_ed8 delta=0.45100: BH eps=6.43 (561s)
A_q03_ed8 delta=0.44900: dispersed eps=7.81 (576s)
A_q03_ed8 delta=0.45000: BH eps=7.23 (564s)
A_q03_ed8 delta=0.44950: BH eps=9.08 (598s)
A_q03_ed8 delta=0.44925: dispersed eps=8.15 (577s)
RESULT A_q03_ed8: delta_c = 0.44938 +- 0.00012 (47.9 min)
```

```
B_n6_ed2 delta=0.51350: BH eps=4.54 (1433s)
B_n6_ed2 delta=0.50525: dispersed eps=5.51 (1441s)
B_n6_ed2 delta=0.50938: BH eps=5.28 (1434s)
B_n6_ed2 delta=0.50731: dispersed eps=6.58 (1464s)
```

```
B_n6_ed2_fin delta=0.50800: BH eps=5.99
B_n6_ed2_fin delta=0.50869: BH eps=5.54
B_n6_ed2_fin delta=0.50754: BH eps=6.96
B_n6_ed2_fin delta=0.50777: BH eps=6.27
RESULT B_n6_ed2_fin: delta_c = 0.50743 +- 0.00011 (48.5 min)
```

B.5 Kept-sector memory curves

```
# SCRIPT: PBH-NU-MEMORY-CURVE  kept-sector survival (memory form) curves for the figures; same object as WB-
    STEVE-07
import numpy as np, json
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
kn,kw=roots_legendre(36); kmin,kmax=0.25,9.0; ks=0.5*(kmax-kmin)*(kn+1)+kmin; wk=0.5*(kmax-kmin)*kw
wE=wk*ks**2*np.exp(-ks**2/2); wE/=wE.sum()
LM=60; M=np.zeros((LM+1,LM+1))
for l in range(1,LM+1): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
tau0=0.02; eg=np.linspace(tau0,6.0,1200); n=LM+1; P=np.zeros(n); P[0]=1
def surv(k,ed,p):
    def rhs(t,v):
        K=0.0 if ed is None else ed**p/t**(p+1)
        x,y=v[:n],v[n:]
        return np.concatenate([-K*(1-P)*x+k*(M@y),-K*(1-P)*y-k*(M@x)])
    v0=np.zeros(2*n); v0[0]=1
    s=solve_ivp(rhs,(tau0,eg[-1]),v0,method='BDF',t_eval=eg,rtol=1e-9,atol=1e-12)
    return s.y[0]**2+s.y[n]**2
f=lambda ed,p: sum(w_*surv(kk,ed,p) for kk,w_ in zip(ks,wE))
i17,i40=np.searchsorted(eg,1.7),np.searchsorted(eg,4.0); w=0.890; Ai=f(None,3)
out={}
for p,eds in [(3,[0.25,0.5,0.75,1,1.5,2,3,4,6,8]),(6,[0.5,1,1.5,2,3,4])]:
    for ed in eds:
        g=(1-f(ed,p))/(1-Ai); out['%d_%g'%(p,ed)]=dict(first_max=float(g[i17]),two_epoch=float(w*g[i17]+(1-w)*
     g[i40]))
        print(p,ed,round(g[i17],3),round(w*g[i17]+(1-w)*g[i40],3),flush=True)
json.dump(out,open('memory_curve.json','w'))
```

```
3 0.25 0.941 0.947
3 0.5 0.86 0.875
3 0.75 0.768 0.792
3 1 0.671 0.705
3 1.5 0.489 0.542
3 2 0.344 0.411
3 3 0.168 0.249
```

```
3 4 0.087 0.169
3 6 0.03 0.095
3 8 0.013 0.059
6 0.5 0.862 0.877
6 1 0.623 0.662
6 1.5 0.352 0.42
6 2 0.159 0.246
6 3 0.026 0.118
6 4 0.005 0.079
```

B.6 The kept-sector readings against McCormick's curve and the history prediction

```
# SCRIPT: WB-STEVE-03-PAPER-OBJECTS
# D2 (memory survival of the kept sector under the time-dependent relaxing operator) and D3 (energy that has
    crossed into the
# hidden sector) as predictors of Steve's G(eps_dec).  Kill conditions are those stated in WB-STEVE-02-LINEAR
    (D2, D3, H1 and
# opposite runs); restated:
#  D2  g = (1-A)/(1-A_inst), A = population of the kept sector l=0 at readout epoch eps* in {1.7, 4.0} (the
    two epochs Steve's
#      own numbers give), no fitted parameter.  Dies if max_eps |g - G| >= 0.10 at BOTH epochs.
#  D3  g = X/X_inst, X = sum_{l>=2} |F_l|^2/(2l+1) summed over the Gaussian perturbation (weights k^2 zeta^2),
     same epochs, same kill.
#  Opposite runs: kept sector = l in {0,1} (flux kept), and k r_m fixed at 2 instead of the profile average.
#  Fitted-epoch scan is diagnostic only (a fitted epoch is not a pass).
import numpy as np
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
import wb_lin
Gs = {0.25: 0.960, 0.5: 0.868, 1.0: 0.684, 2.0: 0.431, 4.0: 0.190, 8.0: 0.063}
kn, kw = roots_legendre(36); kmin, kmax = 0.25, 9.0; ks = 0.5*(kmax-kmin)*(kn+1)+kmin; wk = 0.5*(kmax-kmin)*kw
wE = wk*ks**2*np.exp(-ks**2/2)          # energy weights k^2 zeta(k)^2 of the Gaussian perturbation
wE /= wE.sum()
LM = 60; a = lambda l: l/np.sqrt(4.0*l*l-1)
M = np.zeros((LM+1, LM+1))
for l in range(1, LM+1): M[l, l-1] = M[l-1, l] = a(l)
tau0 = 0.02; eg = np.linspace(tau0, 6.0, 1200)

def survival(k, ed, keep):
    """population of kept sector (keep=1: l=0; keep=2: l=0,1) vs tau, under dv/dtau = -(i k M + Kc (1-P)) v, v
     (tau0)=|0>."""
    n = LM+1; P = np.zeros(n); P[:keep] = 1
    def rhs(t, v):
        K = 0.0 if ed is None else ed**3/t**4
        x, y = v[:n], v[n:]
        return np.concatenate([-K*(1-P)*x+k*(M @ y), -K*(1-P)*y-k*(M @ x)])
    v0 = np.zeros(2*n); v0[0] = 1
    sol = solve_ivp(rhs, (tau0, eg[-1]), v0, method='BDF', t_eval=eg, rtol=1e-9, atol=1e-12)
    pop = sol.y[:keep]**2+sol.y[n:n+keep]**2
    return np.sum(pop, axis=0)

def avg_surv(ed, keep, single=None):
    if single is not None: return survival(single, ed, keep)
    out = np.zeros_like(eg)
    for kk, w_ in zip(ks, wE): out += w_*survival(kk, ed, keep)
    return out

res = {}
for keep, tag in [(1, 'kept=l0'), (2, 'kept=l0,1')]:
    for single, stag in [(None, 'profile-averaged'), (2.0, 'k r_m=2')]:
        Ai = avg_surv(None, keep, single); curves = {}
        for ed in Gs:
            A = avg_surv(ed, keep, single); curves[ed] = (1-A)/(1-Ai)
        res[(tag, stag)] = curves
```

```
        for es in [1.7, 4.0]:
            i = np.searchsorted(eg, es); err = {ed: curves[ed][i]-Gs[ed] for ed in Gs}
            print('D2 %-10s %-17s eps*=%.1f : g = %s  max|err| = %.3f' % (tag, stag, es, np.round([curves[ed][
    i] for ed in Gs], 3), max(abs(v) for v in err.values())))
        # fitted-epoch scan (diagnostic only)
        best = min(((max(abs(curves[ed][i]-Gs[ed]) for ed in Gs), eg[i]) for i in range(40, len(eg), 5)))
        print('   fitted-epoch scan (diagnostic only): best eps* = %.2f with max|err| = %.3f' % (best[1], best
    [0]))
print('Steve G :', [Gs[e] for e in Gs])

# D3 from the full linear solver
def hidden_energy(ed, Rn=0.409):
    X = np.zeros_like(eg)
    for kk, w_ in zip(ks, wE):
        _, Y = wb_lin.run_mode(kk, tau0, eg, Rn, ed)
        F = Y[4:]; ls = np.arange(2, 2+F.shape[0])
        X += w_*np.sum(F**2/(2*ls[:, None]+1), axis=0)
    return X
Xi = hidden_energy(None); D3 = {}
for ed in Gs:
    D3[ed] = hidden_energy(ed)/Xi
for es in [1.7, 4.0]:
    i = np.searchsorted(eg, es); print('D3 eps*=%.1f : g = %s  max|err| = %.3f' % (es, np.round([D3[ed][i] for
     ed in Gs], 3), max(abs(D3[ed][i]-Gs[ed]) for ed in Gs)))
best = min(((max(abs(D3[ed][i]-Gs[ed]) for ed in Gs), eg[i]) for i in range(40, len(eg), 5)))
print('D3 fitted-epoch scan (diagnostic only): best eps* = %.2f with max|err| = %.3f' % (best[1], best[0]))
```

```
D2 kept=l0    profile-averaged  eps*=1.7 : g = [0.941 0.86  0.671 0.344 0.087 0.013]  max|err| = 0.103
D2 kept=l0    profile-averaged  eps*=4.0 : g = [0.997 0.994 0.987 0.959 0.83  0.429]  max|err| = 0.640
   fitted-epoch scan (diagnostic only): best eps* = 1.87 with max|err| = 0.065
/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/a_wb/wb_steve_03_paper_objects.py
    :46: RuntimeWarning: invalid value encountered in divide
  A = avg_surv(ed, keep, single); curves[ed] = (1-A)/(1-Ai)
D2 kept=l0    k r_m=2           eps*=1.7 : g = [1.    0.955 0.772 0.393 0.097 0.014]  max|err| = 0.093
D2 kept=l0    k r_m=2           eps*=4.0 : g = [1.001 1.007 1.016 1.006 1.008 0.587]  max|err| = 0.818
   fitted-epoch scan (diagnostic only): best eps* = 1.69 with max|err| = 0.095
D2 kept=l0,1  profile-averaged  eps*=1.7 : g = [0.981 0.922 0.735 0.373 0.09  0.013]  max|err| = 0.100
D2 kept=l0,1  profile-averaged  eps*=4.0 : g = [0.995 0.983 0.941 0.817 0.544 0.217]  max|err| = 0.386
   fitted-epoch scan (diagnostic only): best eps* = 1.84 with max|err| = 0.082
D2 kept=l0,1  k r_m=2           eps*=1.7 : g = [0.986 0.933 0.752 0.385 0.094 0.014]  max|err| = 0.096
D2 kept=l0,1  k r_m=2           eps*=4.0 : g = [0.998 0.993 0.979 0.882 0.584 0.261]  max|err| = 0.451
   fitted-epoch scan (diagnostic only): best eps* = 1.77 with max|err| = 0.089
Steve G : [0.96, 0.868, 0.684, 0.431, 0.19, 0.063]
/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/a_wb/wb_steve_03_paper_objects.py
    :66: RuntimeWarning: invalid value encountered in divide
  D3[ed] = hidden_energy(ed)/Xi
D3 eps*=1.7 : g = [1.006e+00 9.420e-01 6.770e-01 2.620e-01 3.200e-02 1.000e-03]  max|err| = 0.169
D3 eps*=4.0 : g = [1.024 1.071 1.154 1.168 0.872 0.198]  max|err| = 0.737
D3 fitted-epoch scan (diagnostic only): best eps* = 1.92 with max|err| = 0.112
```

```
# SCRIPT: WB-STEVE-07-HISTORY
# Discriminating prediction for the next step.  Table II interpolation cannot predict a DIFFERENT collision
    history, the
# two-epoch memory-survival form can.  kappa = H (eps_dec/eps)^n  with n = 2 and 6 (Steve used n = 3), eps_dec
     = 1 and 2.
# KILL CONDITION (before the run): the test is only worth asking Steve for if the predicted G for n = 2 or n =
     6 differs from the
# n = 3 value by at least 0.10 at some eps_dec; otherwise the memory form is insensitive to history and the
    step is dropped.
import numpy as np
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
kn, kw = roots_legendre(36); kmin, kmax = 0.25, 9.0; ks = 0.5*(kmax-kmin)*(kn+1)+kmin; wk = 0.5*(kmax-kmin)*kw
wE = wk*ks**2*np.exp(-ks**2/2); wE /= wE.sum()
LM = 60; M = np.zeros((LM+1, LM+1))
for l in range(1, LM+1): M[l, l-1] = M[l-1, l] = l/np.sqrt(4.0*l*l-1)
tau0 = 0.02; eg = np.linspace(tau0, 6.0, 1200); n = LM+1; P = np.zeros(n); P[0] = 1
```

```
def surv(k, ed, p):
    def rhs(t, v):
        K = 0.0 if ed is None else ed**p/t**(p+1)       # a*kappa = (1/tau)(eps_dec/tau)^p
        x, y = v[:n], v[n:]
        return np.concatenate([-K*(1-P)*x+k*(M @ y), -K*(1-P)*y-k*(M @ x)])
    v0 = np.zeros(2*n); v0[0] = 1
    sol = solve_ivp(rhs, (tau0, eg[-1]), v0, method='BDF', t_eval=eg, rtol=1e-9, atol=1e-12)
    return sol.y[0]**2+sol.y[n]**2
f = lambda ed, p: sum(w_*surv(kk, ed, p) for kk, w_ in zip(ks, wE))
i17, i40 = np.searchsorted(eg, 1.7), np.searchsorted(eg, 4.0); w = 0.890
Ai = f(None, 3)
res = {}
for p in [2, 3, 6]:
    for ed in [1.0, 2.0]:
        g = (1-f(ed, p))/(1-Ai); res[(p, ed)] = w*g[i17]+(1-w)*g[i40]
        print('n = %d, eps_dec = %.0f : G_pred(two-epoch, w = 0.89) = %.3f  (n=3 Table II value %.3f)' % (p,
     ed, res[(p, ed)], {1.0: 0.684, 2.0: 0.431}[ed]))
dmax = max(abs(res[(p, ed)]-res[(3, ed)]) for p in [2, 6] for ed in [1.0, 2.0])
print('largest change from n=3: %.3f -> %s' % (dmax, 'discriminating' if dmax >= 0.10 else 'DROPPED (
     insensitive)'))
```

```
n = 2, eps_dec = 1 : G_pred(two-epoch, w = 0.89) = 0.730  (n=3 Table II value 0.684)
n = 2, eps_dec = 2 : G_pred(two-epoch, w = 0.89) = 0.511  (n=3 Table II value 0.431)
n = 3, eps_dec = 1 : G_pred(two-epoch, w = 0.89) = 0.705  (n=3 Table II value 0.684)
n = 3, eps_dec = 2 : G_pred(two-epoch, w = 0.89) = 0.411  (n=3 Table II value 0.431)
n = 6, eps_dec = 1 : G_pred(two-epoch, w = 0.89) = 0.662  (n=3 Table II value 0.684)
n = 6, eps_dec = 2 : G_pred(two-epoch, w = 0.89) = 0.246  (n=3 Table II value 0.431)
largest change from n=3: 0.165 -> discriminating
```

```
original wb_steve_07_out.txt written 2026-10-30 13:47:53.348475647 -0500; Test B launched about 14:30 CT the
     same day
```

B.7 Shear and flux links, and which mean of the rates

```
# SCRIPT: WB-STEVE-01-ALGEBRA
# Tests T0-T4 mapping "The Weight Between" v1.1 onto McCormick's free-streaming PBH preprint (algebraic part).
# KILL CONDITIONS (written before any run):
#  T0  Steve-internal consistency (so the targets are trustworthy). Table II: recompute Delta C_l,c/C_l,c from
      the delta_c
#      columns through Eq.(15) delta = C - 3C^2/8; dies (flag only) if any row differs from the printed value
     by > 0.3 pt.
#      Also recompute G = shift/shift_inst for the gradual rows (flag if any differs from the brief's G by >
     0.005).
#  T1  Prop 5 / Def 4 on Steve's operator L = -i k mu - kappa (I-P), P = projector on l=0 (orthonormal
     Legendre basis).
#      H = k mu, E_B = ||P H (1-P)||_F^2 must equal k^2/3 = (1/2)||[H,P]||_F^2 = Var_{|0>}(H) to 1e-12
     relative. Dies otherwise.
#      Connection-energy identity (scalar form, H' = -L = D + iK, D=diag(kappa_l), K = k mu):
#      (1/4)||[H',H'^dag]||^2 = ||[D,K]||^2 = S_2. Dies if off by > 1e-10 relative.
#  T2  Hidden-walk (Schur) formula of the paper: G_B(z) = [z - E - W (z-F)^{-1} W^dag]^{-1} with E=0, W the l
     =0->l>=1 block,
#      F the hidden block of L, must equal the closed form G(z) = g(s)/(1 - kappa g(s)), g(s)=arctan(k/s)/k, s
     =z+kappa,
#      to 1e-8 relative at 50 random complex z (Re s>0). Dies otherwise.
#      Two-level EP: eigenvalues of [[0,-i k/sqrt3],[-i k/sqrt3,-kappa]] coalesce at kappa = 2 k/sqrt3 = 2
     sqrt(E_B). Dies otherwise.
#  T3  Ladder on Steve's single-rate model: all gaps equal kappa, so S_0^2 = S_-2 S_2 and S_-1^2 = S_-2 S_0
     must be EQUALITIES
#      (1e-12). Dies otherwise.  For the energy-dependent (multigroup) rates the same inequalities must be
     strict.
#  T4  Which rung is the effective single rate of a spread of rates?  Pre-registered claim (to be killed if
     wrong): the paper's
#      ladder says the effective rate is a power mean of the group rates, M_p, p in {-2,-1,0,+1,+2}. PASS only
      if ONE rung p
```

```
#      reproduces BOTH of Steve's calibrated decoupling temperatures, e+- only: 1.6 MeV and with nu-nu:
    0.93-1.0 MeV,
#      each within 15 percent, using only the rate numbers printed in his Appendix C. Opposite run: every
    other p tried.
import numpy as np
from scipy.special import erfc
from scipy.optimize import brentq
rng = np.random.default_rng(3010)
fro2 = lambda X: float(np.real(np.vdot(X, X)))

print('=== T0 Steve-internal consistency ===')
dc_f = 0.49773
rows = [('inst eps0=0.1 Rnu=.409', 0.52832, 9.6), ('dec 0.25', 0.52709, 9.2), ('dec 0.5', 0.52428, 8.3), ('dec
     1', 0.51865, 6.5),
        ('dec 2', 0.51092, 4.0), ('dec 4', 0.50354, 1.8), ('dec 8', 0.49967, 0.6),
        ('Rnu=.10', 0.50547, 2.3), ('Rnu=.20', 0.51297, 4.7), ('Rnu=.30', 0.52027, 7.0), ('Rnu=.488', 0.53379,
     11.5),
        ('alpha .3', 0.47747, 9.3), ('alpha .5', 0.49682, 9.4), ('alpha 2', 0.56342, 9.9), ('alpha 4',
    0.59734, 10.0)]
Cl = lambda d: (4/3)*(1-np.sqrt(1-1.5*d))
fl = {'alpha .3': 0.44805, 'alpha .5': 0.46687, 'alpha 2': 0.53274, 'alpha 4': 0.56819}
for nm, dn, printed in rows:
    df = fl.get(nm, dc_f); s = 100*(Cl(dn)/Cl(df)-1)
    print('%-24s Delta C_l,c = %5.2f %%  printed %5.1f  diff %+5.2f %s' % (nm, s, printed, s-printed, 'FLAG'
     if abs(s-printed) > 0.3 else ''))
sh = {0.25: 5.90, 0.5: 5.33, 1: 4.20, 2: 2.65, 4: 1.17, 8: 0.39}; inst = 6.15
Gbr = {0.25: .959, .5: .867, 1: .683, 2: .431, 4: .190, 8: .063}
G = {e: (v)/inst for e, v in sh.items()}
# use the delta_c columns directly (more digits than the percent column)
dcs = {0.25: 0.52709, .5: 0.52428, 1: 0.51865, 2: 0.51092, 4: 0.50354, 8: 0.49967}
G = {e: (dcs[e]-dc_f)/(0.52832-dc_f) for e in dcs}
for e in G: print('eps_dec %5.2f  G(from delta_c) = %.3f   brief %.3f   diff %+.3f %s' % (e, G[e], Gbr[e], G[e
    ]-Gbr[e], 'FLAG' if abs(G[e]-Gbr[e]) > .005 else ''))
Rv = np.array([.1, .2, .3, .409, .488]); dC = np.array([2.3, 4.7, 7.0, 9.6, 11.5]); dd = np.array([1.56, 3.06,
     4.53, 6.15, 7.24])
print('shift/R_nu  C_l: ', np.round(dC/Rv, 2), ' spread %.1f %%' % (100*np.std(dC/Rv)/np.mean(dC/Rv)))
print('shift/R_nu  delta_c: ', np.round(dd/Rv, 2), ' spread %.1f %%' % (100*np.std(dd/Rv)/np.mean(dd/Rv)))
# Table III slopes and ratios
x = np.array([1e-1, 3e-2, 1e-2, 3e-3, 1e-3, 3e-4, 1e-4]); pf = np.array([.679, .497, .390, .288, .215, .151,
    .105]); fs = np.array([.799, .590, .470, .343, .255, .178, .123])
print('Table III ratios', np.round(fs/pf, 3), ' slopes', np.polyfit(np.log(x), np.log(pf), 1)[0].round(3), np.
    polyfit(np.log(x), np.log(fs), 1)[0].round(3))
# beta^1.18 rule: Gaussian C_l, erfc(C/(sqrt2 sigma))
for b in [1e-9, 1e-12, 1e-15]:
    sig = 0.662/np.sqrt(2)/brentq(lambda y: np.log(erfc(y))-np.log(b), 0.5, 10)
    bn = erfc(0.726/(np.sqrt(2)*sig)); print('beta_pf %.0e -> beta_nu/beta_pf = %.4f (Steve:
    0.021/0.0052/0.0013), exponent %.3f' % (b, bn/b, np.log(bn)/np.log(b)))
print('coincidence list for the 1.19 mass ratio: (1-Rnu)^(-1/3) = %.4f ; (delta_c ratio)^3 = %.4f ; (C_l ratio
    )^2 = %.4f ; Table III 1.17-1.20' %
     ((1-.409)**(-1/3), (0.52832/0.49773)**3, (Cl(0.52832)/Cl(0.49773))**2))
print('Rnu-dependence of a power law (1-Rnu)^-x needed for the C_l shift: x = %.3f at .409, %.3f at .488, %.3f
     at .1 (not constant -> rejected)' %
     tuple(np.log(Cl(d)/Cl(dc_f))/(-np.log(1-r)) for d, r in [(0.52832, .409), (0.53379, .488), (0.50547, .1)
    ]))

print('\n=== T1 Prop 5 / connection energy on Steve operator ===')
lmax = 80; a = lambda l: l/np.sqrt(4.0*l*l-1)
M = np.zeros((lmax+1, lmax+1))
for l in range(1, lmax+1): M[l, l-1] = M[l-1, l] = a(l)
k, kap = 1.7, 0.9
P = np.zeros_like(M); P[0, 0] = 1; I = np.eye(lmax+1)
H = k*M
EB = fro2(P @ H @ (I-P)); comm = 0.5*fro2(H @ P - P @ H); var = (H @ H)[0, 0]-(H)[0, 0]**2
print('E_B = %.15f  k^2/3 = %.15f  (1/2)||[H,P]||^2 = %.15f  Var_0(H) = %.15f' % (EB, k*k/3, comm, var))
assert abs(EB/(k*k/3)-1) < 1e-12 and abs(comm/EB-1) < 1e-12 and abs(var/EB-1) < 1e-12
Dg = np.diag([0.0]+[kap]*lmax); Hp = Dg+1j*H
lhs = 0.25*fro2(Hp @ Hp.conj().T-Hp.conj().T @ Hp); rhs = fro2(Dg @ H-H @ Dg)
```

```
print('(1/4)||[H,H^dag]||^2 = %.12f   ||[D,K]||^2 = %.12f   kappa^2 k^2 * 2/3 = %.12f' % (lhs, rhs, kap**2*k*k
    *2/3))
assert abs(lhs/rhs-1) < 1e-10
print('T1 PASS: sqrt(E_B)/k = %.6f = c_s = 1/sqrt3 (= <mu^2>^(1/2), the radiation sound speed)' % (np.sqrt(EB)
    /k))

print('\n=== T2 hidden-walk (Schur) formula vs arctan closed form ===')
lm = 600; Mb = np.zeros((lm+1, lm+1))
for l in range(1, lm+1): Mb[l, l-1] = Mb[l-1, l] = a(l)
Lm = -1j*k*Mb-kap*(np.eye(lm+1)-np.diag([1.0]+[0]*lm))
W = Lm[0:1, 1:]; F = Lm[1:, 1:]; Wd = Lm[1:, 0:1]
worst = 0
for _ in range(50):
    z = complex(rng.uniform(0.2, 2), rng.uniform(-3, 3))
    GB = 1/(z-0-(W @ np.linalg.solve(z*np.eye(lm)-F, Wd))[0, 0])      # paper formula with E=0
    s = z+kap; g = np.arctan(k/s)/k; Gc = g/(1-kap*g)
    full = np.linalg.inv(z*np.eye(lm+1)-Lm)[0, 0]
    worst = max(worst, abs(GB-Gc)/abs(Gc), abs(full-Gc)/abs(Gc))
print('worst relative difference (Schur vs closed form vs full inverse): %.2e' % worst); assert worst < 1e-8
for kp in [0.5, 2*k/np.sqrt(3)-1e-3, 2*k/np.sqrt(3), 2*k/np.sqrt(3)+1e-3, 3.0]:
    ev = np.linalg.eigvals(np.array([[0, -1j*k/np.sqrt(3)], [-1j*k/np.sqrt(3), -kp]]))
    print('  kappa = %.4f  two-level eigenvalue separation %.2e' % (kp, abs(ev[0]-ev[1])))
print('T2 PASS. EP at kappa = 2 sqrt(E_B) = %.4f' % (2*np.sqrt(EB)))

print('\n=== T3 ladder on the single-rate model ===')
Kc = k*M; gaps = np.zeros((lmax+1, lmax+1)); gaps[0, 1:] = kap; gaps[1:, 0] = kap   # only the cut edge has a
    gap kappa; hidden-hidden gaps are 0
def Sk(kk, gp):
    m = (np.abs(Kc) > 0) & (gp > 0); return float(np.sum(np.abs(Kc[m])**2*gp[m]**kk))
S = {kk: Sk(kk, gaps) for kk in [-2, -1, 0, 1, 2]}
print('S_k:', {kk: round(v, 6) for kk, v in S.items()})
print('equality checks: S0^2/(S-2 S2) = %.12f ; S-1^2/(S-2 S0) = %.12f' % (S[0]**2/(S[-2]*S[2]), S[-1]**2/(S
    [-2]*S[0])))
assert abs(S[0]**2/(S[-2]*S[2])-1) < 1e-12 and abs(S[-1]**2/(S[-2]*S[0])-1) < 1e-12
print('rung +1 for this model: S_1 = E_B*2*kappa (counting both orderings) = %.6f' % S[1])

print('\n=== T4 which rung is the effective rate? (Steve Appendix C numbers only) ===')
# Gamma = c G_F^2 T^4 E.  Calibrate with the unblocked E=T coefficients: 0.506 -> 1.9 MeV (nu_e), 0.108 -> 3.1
     MeV (nu_mu,tau) : T_dec = 1.9 (0.506/C)^(1/3)
def Tdec(C): return 1.9*(0.506/C)**(1/3)
print('sanity: C=0.108 -> %.2f MeV (Steve 3.1)' % Tdec(0.108))
# energy groups: six groups x=0.4..20, edges uniform in ln x (Steve App. C), energy weight x^4/(e^x+1) dlnx, c
    (x) interpolated linearly in ln x between his endpoints
edges = np.linspace(np.log(0.4), np.log(20), 7); xs = np.exp(0.5*(edges[1:]+edges[:-1]))
w = xs**4/(np.exp(xs)+1)*np.diff(edges); w /= w.sum()
def cfun(x, c0, c1): return c0+(c1-c0)*np.log(x/0.5)/np.log(20/0.5)
flav = {'nu_e': dict(wt=1/3, ee=(0.34, 0.47)), 'nu_mutau': dict(wt=2/3, ee=(0.073, 0.101))}
cnn = (0.49, 0.68)
def rates(fl, withnn):
    r = xs*cfun(xs, *flav[fl]['ee']);
    if withnn: r = r+xs*cfun(xs, *cnn)       # anisotropy (l>=2) damped by e+- plus nu-nu
    return r
def pmean(r, wt, p):
    if p == 0: return np.exp(np.sum(wt*np.log(r)))
    return np.sum(wt*r**p)**(1/p)
print('group energy weights', np.round(w, 3))
print('model check: single-group at x=3.15 (e+- only) T_dec: nu_e %.2f MeV (Steve 1.3), nu_mutau %.2f MeV (
    Steve 2.3)' %
      (Tdec(3.15*cfun(3.15, *flav['nu_e']['ee'])), Tdec(3.15*cfun(3.15, *flav['nu_mutau']['ee']))))
res = {}
for withnn in [False, True]:
    for p in [-2, -1, 0, 1, 2]:
        # level 1: power mean over energy groups within a flavour; level 2: same-p mean over flavours (flavour
     weights 1/3, 2/3)
        Cf = [pmean(rates(f, withnn), w, p) for f in flav]; wf = np.array([flav[f]['wt'] for f in flav])
        Ceff = pmean(np.array(Cf), wf, p); res[(withnn, p)] = Tdec(Ceff)
for p in [-2, -1, 0, 1, 2]:
    t1, t2 = res[(False, p)], res[(True, p)]
```

```
    ok = abs(t1/1.6-1) < .15 and (0.93*0.85 <= t2 <= 1.0*1.15)
    print('rung p=%+d : T_dec(e+- only) = %.2f MeV (target 1.6)   T_dec(with nu-nu) = %.2f MeV (target
     0.93-1.0)   both within 15%%: %s' % (p, t1, t2, ok))
print('ratio of masses (T^-2) e+- only -> with nu-nu, rung +1: %.2f, rung -1: %.2f (Steve: 2.5-3)' % ((res[(
     False, 1)]/res[(True, 1)])**2, (res[(False, -1)]/res[(True, -1)])**2))
```

```
=== T0 Steve-internal consistency ===
inst eps0=0.1 Rnu=.409   Delta C_l,c =  9.64 %  printed   9.6  diff +0.04
dec 0.25                 Delta C_l,c =  9.23 %  printed   9.2  diff +0.03
dec 0.5                  Delta C_l,c =  8.31 %  printed   8.3  diff +0.01
dec 1                    Delta C_l,c =  6.48 %  printed   6.5  diff -0.02
dec 2                    Delta C_l,c =  4.04 %  printed   4.0  diff +0.04
dec 4                    Delta C_l,c =  1.76 %  printed   1.8  diff -0.04
dec 8                    Delta C_l,c =  0.58 %  printed   0.6  diff -0.02
Rnu=.10                  Delta C_l,c =  2.35 %  printed   2.3  diff +0.05
Rnu=.20                  Delta C_l,c =  4.68 %  printed   4.7  diff -0.02
Rnu=.30                  Delta C_l,c =  7.00 %  printed   7.0  diff +0.00
Rnu=.488                 Delta C_l,c = 11.47 %  printed  11.5  diff -0.03
alpha .3                 Delta C_l,c =  9.34 %  printed   9.3  diff +0.04
alpha .5                 Delta C_l,c =  9.43 %  printed   9.4  diff +0.03
alpha 2                  Delta C_l,c =  9.91 %  printed   9.9  diff +0.01
alpha 4                  Delta C_l,c = 10.05 %  printed  10.0  diff +0.05
eps_dec  0.25  G(from delta_c) = 0.960   brief 0.959   diff +0.001
eps_dec  0.50  G(from delta_c) = 0.868   brief 0.867   diff +0.001
eps_dec  1.00  G(from delta_c) = 0.684   brief 0.683   diff +0.001
eps_dec  2.00  G(from delta_c) = 0.431   brief 0.431   diff +0.000
eps_dec  4.00  G(from delta_c) = 0.190   brief 0.190   diff -0.000
eps_dec  8.00  G(from delta_c) = 0.063   brief 0.063   diff +0.000
shift/R_nu  C_l:  [23.   23.5  23.33 23.47 23.57]  spread 0.9 %
shift/R_nu  delta_c:  [15.6  15.3  15.1  15.04 14.84]  spread 1.7 %
Table III ratios [1.177 1.187 1.205 1.191 1.186 1.179 1.171]  slopes 0.266 0.267
beta_pf 1e-09 -> beta_nu/beta_pf = 0.0208 (Steve: 0.021/0.0052/0.0013), exponent 1.187
beta_pf 1e-12 -> beta_nu/beta_pf = 0.0053 (Steve: 0.021/0.0052/0.0013), exponent 1.190
beta_pf 1e-15 -> beta_nu/beta_pf = 0.0013 (Steve: 0.021/0.0052/0.0013), exponent 1.192
coincidence list for the 1.19 mass ratio: (1-Rnu)^(-1/3) = 1.1916 ; (delta_c ratio)^3 = 1.1959 ; (C_l ratio)^2
      = 1.2020 ; Table III 1.17-1.20
Rnu-dependence of a power law (1-Rnu)^-x needed for the C_l shift: x = 0.175 at .409, 0.162 at .488, 0.220 at
     .1 (not constant -> rejected)

=== T1 Prop 5 / connection energy on Steve operator ===
E_B = 0.963333333333333  k^2/3 = 0.963333333333333  (1/2)||[H,P]||^2 = 0.963333333333333  Var_0(H) =
     0.963333333333333
(1/4)||[H,H^dag]||^2 = 1.560600000000   ||[D,K]||^2 = 1.560600000000   kappa^2 k^2 * 2/3 = 1.560600000000
T1 PASS: sqrt(E_B)/k = 0.577350 = c_s = 1/sqrt3 (= <mu^2>^(1/2), the radiation sound speed)

=== T2 hidden-walk (Schur) formula vs arctan closed form ===
worst relative difference (Schur vs closed form vs full inverse): 5.29e-16
  kappa = 0.5000  two-level eigenvalue separation 1.90e+00
  kappa = 1.9620  two-level eigenvalue separation 6.26e-02
  kappa = 1.9630  two-level eigenvalue separation 4.18e-08
  kappa = 1.9640  two-level eigenvalue separation 6.27e-02
  kappa = 3.0000  two-level eigenvalue separation 2.27e+00
T2 PASS. EP at kappa = 2 sqrt(E_B) = 1.9630

=== T3 ladder on the single-rate model ===
S_k: {-2: 2.378601, -1: 2.140741, 0: 1.926667, 1: 1.734, 2: 1.5606}
equality checks: S0^2/(S-2 S2) = 1.000000000000 ; S-1^2/(S-2 S0) = 1.000000000000
rung +1 for this model: S_1 = E_B*2*kappa (counting both orderings) = 1.734000

=== T4 which rung is the effective rate? (Steve Appendix C numbers only) ===
sanity: C=0.108 -> 3.18 MeV (Steve 3.1)
group energy weights [0.004 0.038 0.229 0.528 0.199 0.003]
model check: single-group at x=3.15 (e+- only) T_dec: nu_e 1.40 MeV (Steve 1.3), nu_mutau 2.33 MeV (Steve 2.3)
rung p=-2 : T_dec(e+- only) = 2.35 MeV (target 1.6)   T_dec(with nu-nu) = 1.22 MeV (target 0.93-1.0)   both
     within 15%: False
rung p=-1 : T_dec(e+- only) = 2.12 MeV (target 1.6)   T_dec(with nu-nu) = 1.14 MeV (target 0.93-1.0)   both
     within 15%: False
```

```
rung p=+0 : T_dec(e+- only) = 1.86 MeV (target 1.6)   T_dec(with nu-nu) = 1.07 MeV (target 0.93-1.0)   both
    within 15%: False
rung p=+1 : T_dec(e+- only) = 1.62 MeV (target 1.6)   T_dec(with nu-nu) = 1.02 MeV (target 0.93-1.0)   both
    within 15%: True
rung p=+2 : T_dec(e+- only) = 1.44 MeV (target 1.6)   T_dec(with nu-nu) = 0.97 MeV (target 0.93-1.0)   both
    within 15%: True
ratio of masses (T^-2) e+- only -> with nu-nu, rung +1: 2.53, rung -1: 3.44 (Steve: 2.5-3)
```

```
# SCRIPT: WB-STEVE-04-CHANNELS
# (i) Non-normality channels of a collision operator with different rates per multipole (Steve open item (c)),
     via the paper's
#     connection-energy identity.  (ii) Item (e): neutrinos as a self-interacting fluid that exchanges NO
    momentum with the plasma.
# KILL CONDITIONS (written before the run):
#  C1  For L = -i k mu - diag(kappa_l), S_2 = ||[D,K]||^2 = (1/4)||[H',H'^dag]||^2 splits over Legendre edges
    l<->l+1 as
#      sum_l 2 k^2 a_{l+1}^2 (kappa_l - kappa_{l+1})^2, a_l^2 = l^2/(4l^2-1).  Dies if off by > 1e-10 relative
    .
#      Consequence (derived): the relaxation-time model (all kappa_l equal for l>=1) puts ALL non-normality on
     edge 0-1; any
#      model with kappa_1 != kappa_2 puts weight on edge 1-2.
#  E1  Two perfect fluids with equal sound speed and no momentum exchange are one fluid: in pure radiation (p
    = e/3) the shift of
#      the first maximum must vanish as the neutrino self-interaction rate K0 = kappa/H -> infinity.
    Prediction from the paper's
#      rung -1 (viscosity weight of the 1-2 edge, S_-1 = (4/15) k^2/kappa): the shift times K0 tends to a
    constant.
#      Dies if (a) shift does not decrease monotonically with K0, or (b) shift*K0 differs by more than 15
    percent between
#      K0 = 100 and K0 = 1000.  Opposite run: the same operator with the drag rate on (flux relaxed to the
    plasma) instead.
import numpy as np
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre
fro2 = lambda X: float(np.real(np.vdot(X, X)))
lmax = 60; a2 = lambda l: l*l/(4.0*l*l-1)
M = np.zeros((lmax+1, lmax+1))
for l in range(1, lmax+1): M[l, l-1] = M[l-1, l] = np.sqrt(a2(l))
print('=== C1 non-normality channels ===')
k = 1.3
for name, kap in [('relaxation time (all l>=1 equal) kappa_e=1', [0]+[1.0]*lmax),
                  ('nu_e: flux damped by e+- (1), stress by e+- + nu-nu (1+1.5)', [0, 1.0]+[2.5]*(lmax-1)),
                  ('nu_mu,tau: flux (1), stress (1+7)', [0, 1.0]+[8.0]*(lmax-1))]:
    D = np.diag(kap); Hp = D+1j*k*M
    S2 = fro2(D @ (k*M)-(k*M) @ D); chk = 0.25*fro2(Hp @ Hp.conj().T-Hp.conj().T @ Hp)
    edges = [2*k*k*a2(l+1)*(kap[l]-kap[l+1])**2 for l in range(0, 6)]
    tot = sum(2*k*k*a2(l+1)*(kap[l]-kap[l+1])**2 for l in range(0, lmax))
    print('%-62s S_2 = %.6f (commutator form %.6f, edge sum %.6f)' % (name, S2, chk, tot))
    assert abs(S2/chk-1) < 1e-10 and abs(S2/tot-1) < 1e-10
    print('    edge energies 0-1, 1-2, 2-3, 3-4: %s ; share on edge 1-2: %.1f %%' % (np.round(edges[:4], 4),
     100*edges[1]/tot))
wt = np.array([1/3, 2/3]); e12 = np.array([2.25*(4/15), 49*(4/15)]); e01 = np.array([1/3, 1/3])
print('flavour-weighted ratio (stress edge)/(flux edge) in units of kappa_e^2 k^2 : %.2f' % (np.sum(wt*e12)/np
    .sum(wt*e01)))
print('E_B for the cut P0 (plasma-coupled monopole): k^2/3 ; for the cut P01 (flux kept, i.e. nu as own fluid)
    : 4 k^2/15 ; ratio %.4f' % ((4/15)/(1/3)))
E_B01 = fro2(np.diag([1, 1]+[0]*(lmax-1)) @ (k*M) @ np.diag([0, 0]+[1]*(lmax-1)))
print('numerical E_B(P01) = %.10f ; 4k^2/15 = %.10f' % (E_B01, 4*k*k/15))

print('\n=== E1 self-interacting neutrinos with no momentum exchange, pure radiation (first-max shift of Delta
    ) ===')
LM = 40
def run(k, Rn, K0, drag, tau0=0.02, tg=None):
    n = 4+(LM-1); LS = np.arange(3, LM+1)
    def rhs(t, y):
        dg, tg_, dn, tn = y[:4]; F = np.zeros(LM+2); F[2:LM+1] = y[4:]; F[LM+1] = (2*LM+1)/(k*t)*F[LM]-F[LM-1]
        th = (1-Rn)*tg_+Rn*tn; de = (1-Rn)*dg+Rn*dn; sg = F[2]/2; Dl = de+4*th/(k*k*t)
        phi = -1.5*Dl/(k*t)**2; psi = phi-6*Rn*sg/(k*t)**2; dphi = -psi/t+2*th/(k*k*t*t)
```

```
        K = K0/t; dy = np.empty_like(y)               # a*kappa = K0 * (a H) = K0 / tau
        Kd = K if drag else 0.0
        dy[0] = -4/3*tg_+4*dphi
        dy[1] = k*k*dg/4+k*k*psi-Kd*(Rn/(1-Rn))*(tg_-tn)
        dy[2] = -4/3*tn+4*dphi
        dy[3] = k*k*(dn/4-sg)+k*k*psi+Kd*(tg_-tn)
        dy[4] = 8/15*tn-0.6*k*F[3]-K*F[2]
        dy[5:] = k/(2*LS+1)*(LS*F[2:LM]-(LS+1)*F[4:LM+2])-K*F[3:LM+1]
        return dy
    psi0 = 2/3; d0 = -2*psi0-(2/3)*(k*tau0)**2*psi0; t0 = 0.5*k*k*tau0*psi0
    y0 = np.zeros(n); y0[:4] = [d0, t0, d0, t0]
    sol = solve_ivp(rhs, (tau0, tg[-1]), y0, method='BDF', t_eval=tg, rtol=1e-10, atol=1e-13)
    Y = sol.y; th = (1-Rn)*Y[1]+Rn*Y[3]; de = (1-Rn)*Y[0]+Rn*Y[2]
    return de+4*th/(k*k*tg)
def fmax(x, y):
    s = -y
    for i in range(1, len(s)-1):
        if s[i] > s[i-1] and s[i] >= s[i+1]:
            A, B, C = s[i-1], s[i], s[i+1]; off = 0.5*(A-C)/(A-2*B+C); return B-0.25*(A-C)*off
tg = np.linspace(0.02, 8.0, 3000); kk = 1.0
m_pf = fmax(tg, run(kk, 0.0, 0.0, False, tg=tg))
for drag, tag in [(False, 'self-interaction only (no momentum exchange with plasma)'), (True, 'drag on the
     plasma included (Steve relaxation model)')]:
    print(tag)
    prev = None; vals = {}
    for K0 in [3, 10, 30, 100, 300, 1000, 3000]:
        r = fmax(tg, run(kk, 0.409, K0, drag, tg=tg))/m_pf; vals[K0] = 1/r-1
        print('   K0 = %5d : first-max ratio %.5f ; implied threshold rise %.4f %% ; rise*K0 = %.3f' % (K0, r,
      100*vals[K0], 100*vals[K0]*K0))
    if not drag:
        mono = all(vals[a] > vals[b] for a, b in zip([3, 10, 30, 100, 300, 1000], [10, 30, 100, 300, 1000,
     3000]))
        const = abs(vals[1000]*1000/(vals[100]*100)-1)
        print('   E1 (a) monotone decrease: %s ; (b) rise*K0 changes by %.1f %% between K0=100 and 1000 -> %s'
      % (mono, 100*const, 'PASS' if mono and const < .15 else 'DIES'))
```

```
=== C1 non-normality channels ===
relaxation time (all l>=1 equal) kappa_e=1                    S_2 = 1.126667 (commutator form 1.126667, edge
     sum 1.126667)
    edge energies 0-1, 1-2, 2-3, 3-4: [1.1267 0.     0.     0.    ] ; share on edge 1-2: 0.0 %
nu_e: flux damped by e+- (1), stress by e+- + nu-nu (1+1.5)    S_2 = 3.154667 (commutator form 3.154667, edge
     sum 3.154667)
    edge energies 0-1, 1-2, 2-3, 3-4: [1.1267 2.028  0.     0.    ] ; share on edge 1-2: 64.3 %
nu_mu,tau: flux (1), stress (1+7)                             S_2 = 45.292000 (commutator form 45.292000,
     edge sum 45.292000)
    edge energies 0-1, 1-2, 2-3, 3-4: [ 1.1267 44.1653  0.      0.    ] ; share on edge 1-2: 97.5 %
flavour-weighted ratio (stress edge)/(flux edge) in units of kappa_e^2 k^2 : 26.73
E_B for the cut P0 (plasma-coupled monopole): k^2/3 ; for the cut P01 (flux kept, i.e. nu as own fluid): 4 k
     ^2/15 ; ratio 0.8000
numerical E_B(P01) = 0.4506666667 ; 4k^2/15 = 0.4506666667

=== E1 self-interacting neutrinos with no momentum exchange, pure radiation (first-max shift of Delta) ===
self-interaction only (no momentum exchange with plasma)
   K0 =     3 : first-max ratio 0.96022 ; implied threshold rise 4.1424 % ; rise*K0 = 12.427
   K0 =    10 : first-max ratio 0.98310 ; implied threshold rise 1.7192 % ; rise*K0 = 17.192
   K0 =    30 : first-max ratio 0.99370 ; implied threshold rise 0.6345 % ; rise*K0 = 19.035
   K0 =   100 : first-max ratio 0.99804 ; implied threshold rise 0.1969 % ; rise*K0 = 19.688
   K0 =   300 : first-max ratio 0.99934 ; implied threshold rise 0.0662 % ; rise*K0 = 19.849
   K0 =  1000 : first-max ratio 0.99980 ; implied threshold rise 0.0199 % ; rise*K0 = 19.918
   K0 =  3000 : first-max ratio 0.99993 ; implied threshold rise 0.0067 % ; rise*K0 = 20.088
   E1 (a) monotone decrease: True ; (b) rise*K0 changes by 1.2 % between K0=100 and 1000 -> PASS
drag on the plasma included (Steve relaxation model)
   K0 =     3 : first-max ratio 0.95931 ; implied threshold rise 4.2421 % ; rise*K0 = 12.726
   K0 =    10 : first-max ratio 0.98281 ; implied threshold rise 1.7488 % ; rise*K0 = 17.488
   K0 =    30 : first-max ratio 0.99364 ; implied threshold rise 0.6396 % ; rise*K0 = 19.187
   K0 =   100 : first-max ratio 0.99803 ; implied threshold rise 0.1974 % ; rise*K0 = 19.743
   K0 =   300 : first-max ratio 0.99934 ; implied threshold rise 0.0664 % ; rise*K0 = 19.910
   K0 =  1000 : first-max ratio 0.99980 ; implied threshold rise 0.0200 % ; rise*K0 = 20.042
```

```
   K0 =  3000 : first-max ratio 0.99993 ; implied threshold rise 0.0067 % ; rise*K0 = 20.210
```

B.8 The full operator and the truncated wall

```
# SCRIPT: OSL-STEVE-B-EP-FULL-OPERATOR
# Question: Steve's neutrino operator L = -i k mu - kappa (I - P0) (momentum given to plasma, so l=1 is damped
     too).
# Paper-derived claim to test: the two-moment truncation has its exceptional point at kappa_EP = 2 sqrt(E_B) =
     2k/sqrt3
# (critical damping of a pair coupled with weight E_B = k^2/3). Does that EP survive when all l are kept?
# KILL CONDITIONS (written before the run):
# KB1 (truncation control) N=2 EP must come out at 2/sqrt3 = 1.1547 (k=1) to 1e-6, else the code is wrong.
# KB2 the claim 'kappa_EP = 2 sqrt(E_B)' survives in the full operator iff the N=200 value is within 10% of
     1.1547.
# KB3 (opposite / inverse reading) the statement 'above kappa_EP the neutrinos are overdamped (no oscillating
     mode)' survives
#     iff for N=200 every eigenvalue with Re > -kappa/2... is real for kappa > 1.5*kappa_EP; we report the
     least-damped eigenvalue's Im.
import numpy as np
def L(N, kap, k=1.0):
    n = np.arange(1, N); a = n / np.sqrt(4.0 * n**2 - 1.0)       # mu P_n = a_{n+1} P_{n+1} + a_n P_{n-1},
     orthonormal Legendre
    J = np.diag(a, 1) + np.diag(a, -1)
    D = np.eye(N); D[0, 0] = 0.0
    return -1j * k * J - kap * D
def slowest(N, kap):
    ev = np.linalg.eigvals(L(N, kap)); i = np.argmax(ev.real); return ev[i], ev
def ep(N, lo=0.01, hi=6.0):
    # smallest kappa such that the least-damped eigenvalue is real (|Im| < 1e-9) and stays real above it
    ks = np.linspace(lo, hi, 5901); flag = np.array([abs(slowest(N, x)[0].imag) < 1e-7 for x in ks])
    # last kappa at which a non-real least-damped eigenvalue occurs
    bad = np.where(~flag)[0]
    return ks[bad.max() + 1] if len(bad) and bad.max() + 1 < len(ks) else (ks[0] if len(bad) == 0 else np.nan)
print('2/sqrt3 = %.6f' % (2 / 3**0.5))
res = {}
for N in [2, 3, 5, 10, 40, 200]:
    res[N] = ep(N); print('N = %3d : least-damped eigenvalue becomes real (and stays real) for kappa >= %.4f'
     % (N, res[N]))
print('KB1: |EP(N=2) - 2/sqrt3| = %.2e' % abs(res[2] - 2 / 3**0.5))
print('KB2: EP(N=200)/(2/sqrt3) = %.3f' % (res[200] / (2 / 3**0.5)))
for kap in [0.5, 1.0, 1.5, 2.0, 3.0, 6.0]:
    for N in [2, 200]:
        lam, ev = slowest(N, kap); print('kappa=%.1f N=%3d least-damped lambda = %+.4f %+.4fi ; diffusion
     estimate -k^2/(3 kappa) = %+.4f' % (kap, N, lam.real, lam.imag, -1 / (3 * kap)))
```

```
2/sqrt3 = 1.154701
N =   2 : least-damped eigenvalue becomes real (and stays real) for kappa >= 1.1552
N =   3 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.0100
N =   5 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.0100
N =  10 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.7552
N =  40 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.6719
N = 200 : least-damped eigenvalue becomes real (and stays real) for kappa >= 0.6455
KB1: |EP(N=2) - 2/sqrt3| = 5.06e-04
KB2: EP(N=200)/(2/sqrt3) = 0.559
kappa=0.5 N=  2 least-damped lambda = -0.2500 -0.5204i ; diffusion estimate -k^2/(3 kappa) = -0.6667
kappa=0.5 N=200 least-damped lambda = -0.4947 -0.0078i ; diffusion estimate -k^2/(3 kappa) = -0.6667
kappa=1.0 N=  2 least-damped lambda = -0.5000 +0.2887i ; diffusion estimate -k^2/(3 kappa) = -0.3333
kappa=1.0 N=200 least-damped lambda = -0.3579 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.3333
kappa=1.5 N=  2 least-damped lambda = -0.2713 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.2222
kappa=1.5 N=200 least-damped lambda = -0.2291 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.2222
kappa=2.0 N=  2 least-damped lambda = -0.1835 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1667
kappa=2.0 N=200 least-damped lambda = -0.1695 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1667
kappa=3.0 N=  2 least-damped lambda = -0.1156 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1111
kappa=3.0 N=200 least-damped lambda = -0.1119 -0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.1111
kappa=6.0 N=  2 least-damped lambda = -0.0561 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.0556
```

```
kappa=6.0 N=200 least-damped lambda = -0.0557 +0.0000i ; diffusion estimate -k^2/(3 kappa) = -0.0556

[exited with code 0]
```

B.9 The annihilation contest

```
# SCRIPT: OSL-STEVE-C-SPEED-CONTEST-EOS
# Mapping under test (from the paper + Check A): the neutrino rms streaming speed is sqrt(E_B)/k = sqrt(<mu
    ^2>) = 1/sqrt3 (exact) and the
# plasma sound speed is c_s = sqrt(dp/de). In pure radiation they coincide; during e+- annihilation c_s falls
    below 1/sqrt3, so a
# 'speed contest' reading predicts the free-streaming advantage GROWS there. Pre-registered quantitative form
    (no fit):
#   shift_pred(T_H) = 0.15 * R_nu(T_H) * G(eps_dec = T_H/T_dec) * f,  f = (1 - c_s(T_H)) / (1 - 1/sqrt3)   (
    front speed 1 against sound)
#   baseline        = 0.15 * R_nu(T_H) * G(...)                                                          (
    no contrast factor)
# Steve's published envelope: relative shift of delta_c is 5.3% to 6.2% at ALL sample masses 3e5 .. 7e7 Msun (
    Sec. IV D).
# VALIDATION (must pass first): my plasma p/e minimum must be 0.274 +- 0.003 near T = 0.16 MeV; R_nu = 0.488 (
    T>>m_e) and 0.405 (T<<m_e).
# KILL CONDITIONS (written before the run):
# KC0 validation fails -> stop, code not trusted.
# KC1 'contrast enhancement' mapping dies if shift_pred (with f) exceeds 6.5% at any sample mass between 3e6
    and 7e7 Msun (G ~ 1 there).
# KC2 opposite reading (the contrast SHRINKS the effect: f' = 1/f) dies if shift_pred(f') is below 5.3% at any
     sample mass 3e5..7e7.
import numpy as np
from scipy.integrate import quad
me = 0.511
def fd(T, n):  # fermi-dirac integrals for g=4 (e+-), zero chem. potential
    pass
def eP(T):
    f = lambda p, w: p**2 * w(p) / (np.exp(np.sqrt(p**2 + me**2) / T) + 1)
    e = quad(lambda p: p**2 * np.sqrt(p**2 + me**2) / (np.exp(np.sqrt(p**2 + me**2) / T) + 1), 0, 60 * T + 10)
     [0] * 4 / (2 * np.pi**2)
    P = quad(lambda p: p**4 / np.sqrt(p**2 + me**2) / (np.exp(np.sqrt(p**2 + me**2) / T) + 1), 0, 60 * T + 10)
     [0] * 4 / (6 * np.pi**2)
    return e + np.pi**2 / 15 * T**4, P + np.pi**2 / 45 * T**4
def sdens(T): e, P = eP(T); return (e + P) / T
s_hi_coef = 2 * np.pi**2 / 45 * 5.5
def Tnu(T): return (sdens(T) / s_hi_coef)**(1 / 3)
def Rnu(T): enu = 6 * 7 / 8 * np.pi**2 / 30 * Tnu(T)**4; e, P = eP(T); return enu / (enu + e)
def cs2(T): h = 1e-4 * T; (e1, P1), (e2, P2) = eP(T - h), eP(T + h); return (P2 - P1) / (e2 - e1)
Ts = np.geomspace(0.03, 20, 160); pe = np.array([eP(T)[1] / eP(T)[0] for T in Ts]); i = np.argmin(pe)
print('KC0 validation: min p/e = %.4f at T = %.3f MeV (Steve: 0.274 at ~0.16)' % (pe[i], Ts[i]))
print('   R_nu(T=20) = %.4f (Steve 0.488) ; R_nu(T=0.03) = %.4f (Steve 0.405 for Neff=3)' % (Rnu(20.0), Rnu
    (0.03)))
c2 = np.array([cs2(T) for T in Ts]); j = np.argmin(c2); print('   min c_s^2 = dp/de = %.4f (c_s = %.4f) at T =
     %.3f MeV ; 1/3 = 0.3333' % (c2[j], c2[j]**0.5, Ts[j]))
ok0 = abs(pe[i] - 0.274) < 0.003 and abs(Rnu(20.0) - 0.488) < 0.002 and abs(Rnu(0.03) - 0.405) < 0.002
print('KC0 PASS' if ok0 else 'KC0 FAIL')
# horizon temperature from mass: M_H = 1.5e5 (T/MeV)^-2 (g*/10.75)^-1/2 ; g* defined by rho = (pi^2/30) g* T^4
     (plasma + nu)
def gstar(T):
    e, P = eP(T); enu = 6 * 7 / 8 * np.pi**2 / 30 * Tnu(T)**4; return (e + enu) / (np.pi**2 / 30 * T**4)
def TH_of(M):
    from scipy.optimize import brentq
    return brentq(lambda T: 1.5e5 * T**-2 * (gstar(T) / 10.75)**-0.5 - M, 0.02, 30)
Gtab = {0.25: 0.959, 0.5: 0.867, 1: 0.683, 2: 0.431, 4: 0.190, 8: 0.063}
def G(ed):
    x = np.log(list(Gtab.keys())); y = list(Gtab.values()); return float(np.interp(np.log(ed), x, y, left=1.0,
     right=0.063 * (8 / ed)**2))
print('\n  M_H[Msun]   T_H[MeV]  R_nu    c_s     f      0.15 R G   with f     with 1/f')
rows = []
for M in [3e5, 1e6, 3e6, 7e6, 1.6e7, 3e7, 7e7]:
```

```
    T = TH_of(M); R = Rnu(T); c = cs2(T)**0.5; f = (1 - c) / (1 - 3**-0.5); g_ = G(T / 1.0)
    base = 0.15 * R * g_ * 100; rows.append((M, T, base, base * f, base / f))
    print('  %8.1e  %8.3f  %.3f  %.4f  %.3f  %6.2f%%   %6.2f%%   %6.2f%%' % (M, T, R, c, f, base, base * f,
     base / f))
heavy = [r for r in rows if r[0] >= 3e6]
print('KC1: max shift_pred(with f) for M >= 3e6 = %.2f%% (kill if > 6.5%%) -> %s' % (max(r[3] for r in heavy),
      'FAIL' if max(r[3] for r in heavy) > 6.5 else 'PASS'))
print('KC2: min shift_pred(with 1/f) over all = %.2f%% (kill if < 5.3%%) -> %s' % (min(r[4] for r in rows), '
     FAIL' if min(r[4] for r in rows) < 5.3 else 'PASS'))
print('baseline (no f): range %.2f%% - %.2f%% ; envelope 5.3 - 6.2' % (min(r[2] for r in rows), max(r[2] for r
      in rows)))
```

```
KC0 validation: min p/e = 0.2736 at T = 0.167 MeV (Steve: 0.274 at ~0.16)
   R_nu(T=20) = 0.4884 (Steve 0.488) ; R_nu(T=0.03) = 0.4052 (Steve 0.405 for Neff=3)
   min c_s^2 = dp/de = 0.2697 (c_s = 0.5193) at T = 0.131 MeV ; 1/3 = 0.3333
KC0 PASS

  M_H[Msun]   T_H[MeV]  R_nu    c_s     f      0.15 R G   with f     with 1/f
   3.0e+05     0.713  0.483  0.5709  1.015    5.60%     5.68%     5.52%
   1.0e+06     0.398  0.474  0.5591  1.043    6.38%     6.65%     6.11%
   3.0e+06     0.240  0.457  0.5391  1.091    6.85%     7.47%     6.28%
   7.0e+06     0.166  0.439  0.5235  1.128    6.58%     7.42%     5.83%
   1.6e+07     0.118  0.421  0.5205  1.135    6.31%     7.16%     5.56%
   3.0e+07     0.090  0.411  0.5337  1.103    6.17%     6.81%     5.59%
   7.0e+07     0.061  0.406  0.5642  1.031    6.09%     6.28%     5.90%
KC1: max shift_pred(with f) for M >= 3e6 = 7.47% (kill if > 6.5%) -> FAIL
KC2: min shift_pred(with 1/f) over all = 5.52% (kill if < 5.3%) -> PASS
baseline (no f): range 5.60% - 6.85% ; envelope 5.3 - 6.2
```

B.10 Linear first maximum under gradual decoupling

```
# SCRIPT: LIN-FS-01  linear free-streaming + relaxation collisions, radiation era, Newtonian gauge (Ma-
    Bertschinger)
import numpy as np
from scipy.integrate import solve_ivp
LMAX=36
def build(k,Rnu,eps_dec,r_m=1.0):
    n=4+(LMAX-1)  # dg,tg,dn,tn,F2..FLMAX
    iF=lambda l:4+(l-2)
    def A(t):
        M=np.zeros((n,n))
        kt2=(k*t)**2
        # phi, thetaT, deltaT, sigma as linear functionals
        dT=np.zeros(n); dT[0]=1-Rnu; dT[2]=Rnu
        thT=np.zeros(n); thT[1]=1-Rnu; thT[3]=Rnu
        sig=np.zeros(n); sig[iF(2)]=0.5
        phi=-1.5/kt2*(dT+4*thT/(k*k*t))
        psi=phi-6*Rnu*sig/kt2
        phid=-psi/t+2*thT/(t*t*k*k)
        aK=0.0 if eps_dec is None else (1.0/t)*((eps_dec*r_m/t)**3)
        M[0]=4*phid; M[0,1]+=-4/3.
        M[1]=k*k*psi; M[1,0]+=k*k/4
        if Rnu>0:
            c=aK*Rnu/(1-Rnu); M[1,3]+=c; M[1,1]+=-c
        M[2]=4*phid; M[2,3]+=-4/3.
        M[3]=k*k*psi; M[3,2]+=k*k/4; M[3]-=k*k*sig; M[3,3]+=-aK; M[3,1]+=aK
        # F2
        M[iF(2),3]+=8/15.; M[iF(2),iF(3)]+=-0.6*k; M[iF(2),iF(2)]+=-aK
        for l in range(3,LMAX+1):
            r=iF(l)
            M[r,iF(l-1)]+=k*l/(2*l+1.)
            if l<LMAX: M[r,iF(l+1)]+=-k*(l+1)/(2*l+1.)
            else:
                # closure F_{L+1}=(2L+1)/(k t) F_L - F_{L-1}
                M[r,iF(l)]+=-k*(l+1)/(2*l+1.)*((2*l+1)/(k*t)); M[r,iF(l-1)]+=k*(l+1)/(2*l+1.)
```

```
            M[r,r]+=-aK
        return M,dT,thT,phi
    return n,A
def run(k,Rnu,eps_dec,t0,tmax,r_m=1.0,npts=400):
    n,A=build(k,Rnu,eps_dec,r_m)
    zeta=1.0; psi0=2/3.*zeta
    th0=0.5*k*k*t0*psi0
    d0=-2*psi0-(2/3.)*(k*t0)**2*psi0
    y0=np.zeros(n); y0[0]=d0; y0[1]=th0; y0[2]=d0; y0[3]=th0
    f=lambda t,y:A(t)[0]@y
    J=lambda t,y:A(t)[0]
    ts=np.linspace(t0,tmax,npts)
    s=solve_ivp(f,(t0,tmax),y0,method='Radau',jac=J,t_eval=ts,rtol=1e-8,atol=1e-11)
    Y=s.y; out={}
    dT=(1-Rnu)*Y[0]+Rnu*Y[2]; thT=(1-Rnu)*Y[1]+Rnu*Y[3]
    out['t']=s.t
    out['Delta']=dT+4*thT/(k*k*s.t)           # zero-momentum slices
    out['dpl']=dT+4*Y[1]/(k*k*s.t)            # plasma-orthogonal slices
    return out
def firstmax(t,y):
    i=np.where((y[1:-1]>y[:-2])&(y[1:-1]>=y[2:]))[0]
    if len(i)==0: return None
    j=i[0]+1
    # parabolic refinement
    a,b,c=y[j-1],y[j],y[j+1]
    d=0.5*(a-c)/(a-2*b+c); return t[j]+d*(t[1]-t[0]), b-0.25*(a-c)*d
if __name__=="__main__":
    # validation: single mode, nu decoupled from start (instant) vs perfect fluid
    for k in [1.0]:
        for Rnu in [0.409,0.488]:
            pf=run(k,0.0,None,0.01/k,12/k)
            fs=run(k,Rnu,None,0.01/k,12/k)
            for key in ['Delta','dpl']:
                m1=firstmax(pf['t'],pf[key]); m2=firstmax(fs['t'],fs[key])
                print(k,Rnu,key,m1,m2,'ratio',m2[1]/m1[1])
```

```
# SCRIPT: LIN-FS-03  central first-max factor vs gradual decoupling kappa=H(eps_dec/eps)^3 (linear theory,
     Gaussian profile)
import numpy as np, time, sys
from multiprocessing import Pool
from lin import run, firstmax
xk,wk=np.polynomial.legendre.leggauss(28)
kmax=8.0
ks=0.5*kmax*(xk+1); ws=0.5*kmax*wk
T=np.linspace(0.1,4.0,586)
def job(a):
    Rnu,ed,k=a
    r=run(k,Rnu,ed,0.1,4.0,npts=len(T))
    return (-r['Delta'],-r['dpl'])
def centre(Rnu,ed,pool):
    out=pool.map(job,[(Rnu,ed,k) for k in ks])
    D=sum(w*k*k*np.exp(-k*k/4.)*o[0] for o,k,w in zip(out,ks,ws))/(2*np.pi**2)
    P=sum(w*k*k*np.exp(-k*k/4.)*o[1] for o,k,w in zip(out,ks,ws))/(2*np.pi**2)
    return D,P
if __name__=="__main__":
    t0=time.time()
    with Pool(4) as pool:
        pf=centre(0.0,None,pool); mp=[firstmax(T,x)[1] for x in pf]
        res=[]
        for ed in [None,0.25,0.5,1,2,4,8]:
            fs=centre(0.409,ed,pool)
            mf=[firstmax(T,x) for x in fs]
            sh=[100*(mp[i]/mf[i][1]-1) for i in range(2)]
            print("edec",ed,"Delta shift %.3f  dpl shift %.3f  tmax %.3f %.3f  t=%.0f"%(sh[0],sh[1],mf[0][0],
     mf[1][0],time.time()-t0),flush=True)
```

```
edec None Delta shift 5.962  dpl shift 6.663  tmax 1.711 1.744  t=119
```

```
edec 0.25 Delta shift 5.124  dpl shift 5.878  tmax 1.711 1.743  t=182
edec 0.5 Delta shift 3.498  dpl shift 4.233  tmax 1.714 1.740  t=233
edec 1 Delta shift 1.109  dpl shift 1.423  tmax 1.730 1.746  t=280
edec 2 Delta shift -0.285  dpl shift -0.474  tmax 1.748 1.757  t=327
edec 4 Delta shift -0.145  dpl shift -0.197  tmax 1.737 1.739  t=369
edec 8 Delta shift -0.022  dpl shift -0.023  tmax 1.731 1.731  t=411
```

B.11 Rate-ratio and wall readings

```
# SCRIPT: CONN-ENERGY-STEVE-04  corrected scaling k/(aH) = k r_m * eps
import numpy as np
from scipy.optimize import brentq
G=np.array([5.90,5.33,4.20,2.65,1.17,0.39])/6.15; ed=np.array([.25,.5,1,2,4,8])
print('EP crossing (k r_m=2): kappa/H=(edec/eps)^3 equals (2/sqrt3)*2*eps')
print([round(brentq(lambda e:(x/e)**3-(4/np.sqrt(3))*e,1e-3,100),3) for x in ed], ' vs 0.811*edec^0.75 =',np.
    round(0.811*ed**0.75,3))
forms={'wall sqrt(1-3kap^2/4k^2)':lambda kap,k:np.sqrt(np.clip(1-0.75*kap**2/k**2,0,1)),
       'A k/(k+kappa)':lambda kap,k:k/(k+kap),'B exp(-kappa/k)':lambda kap,k:np.exp(-kap/k),'C 1-exp(-k/kappa)
    ':lambda kap,k:1-np.exp(-k/kap)}
for lo in [1.0,0.1]:
  eps=np.linspace(lo,4,6001)
  for kr in [1.0,2.0]:
    for name,f in forms.items():
      for wname,wt in [('uniform',np.ones_like(eps)),('log',1/eps)]:
        pred=np.array([np.sum(f((e/eps)**3,kr*eps)*wt)/np.sum(wt) for e in ed]); err=np.max(np.abs(pred-G))
        print('win %.1f-4 k r_m=%.0f %-26s %-8s %s maxerr %.3f %s'%(lo,kr,name,wname,np.round(pred,3),err,'
    PASS' if err<=0.10 else ''))
```

```
EP crossing (k r_m=2): kappa/H=(edec/eps)^3 equals (2/sqrt3)*2*eps
[0.287, 0.482, 0.811, 1.364, 2.294, 3.859]  vs 0.811*edec^0.75 = [0.287 0.482 0.811 1.364 2.294 3.858]
win 1.0-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   uniform  [1.    1.    0.98  0.746 0.35  0.   ] maxerr 0.315
win 1.0-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   log      [1.    0.999 0.962 0.596 0.223 0.   ] maxerr 0.279
win 1.0-4 k r_m=1 A k/(k+kappa)              uniform  [0.998 0.987 0.92  0.717 0.386 0.105] maxerr 0.286
win 1.0-4 k r_m=1 A k/(k+kappa)              log      [0.997 0.979 0.876 0.609 0.287 0.073] maxerr 0.193
win 1.0-4 k r_m=1 B exp(-kappa/k)            uniform  [0.998 0.987 0.91  0.66  0.283 0.015] maxerr 0.229
win 1.0-4 k r_m=1 B exp(-kappa/k)            log      [0.997 0.978 0.857 0.526 0.188 0.009] maxerr 0.174
win 1.0-4 k r_m=1 C 1-exp(-k/kappa)          uniform  [1.    1.    0.979 0.817 0.479 0.117] maxerr 0.386
win 1.0-4 k r_m=1 C 1-exp(-k/kappa)          log      [1.    1.    0.96  0.707 0.352 0.08 ] maxerr 0.277
win 1.0-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   uniform  [1.    1.    0.995 0.839 0.504 0.016] maxerr 0.408
win 1.0-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   log      [1.    1.    0.991 0.72  0.346 0.009] maxerr 0.308
win 1.0-4 k r_m=2 A k/(k+kappa)              uniform  [0.999 0.993 0.954 0.802 0.503 0.177] maxerr 0.371
win 1.0-4 k r_m=2 A k/(k+kappa)              log      [0.999 0.989 0.927 0.712 0.391 0.124] maxerr 0.282
win 1.0-4 k r_m=2 B exp(-kappa/k)            uniform  [0.999 0.993 0.951 0.762 0.416 0.066] maxerr 0.331
win 1.0-4 k r_m=2 B exp(-kappa/k)            log      [0.999 0.989 0.92  0.648 0.293 0.04 ] maxerr 0.237
win 1.0-4 k r_m=2 C 1-exp(-k/kappa)          uniform  [1.    1.    0.996 0.89  0.613 0.207] maxerr 0.460
win 1.0-4 k r_m=2 C 1-exp(-k/kappa)          log      [1.    1.    0.991 0.812 0.473 0.143] maxerr 0.381
win 0.1-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   uniform  [0.931 0.866 0.757 0.574 0.269 0.   ] maxerr 0.143
win 0.1-4 k r_m=1 wall sqrt(1-3kap^2/4k^2)   log      [0.646 0.505 0.364 0.224 0.084 0.   ] maxerr 0.361
win 0.1-4 k r_m=1 A k/(k+kappa)              uniform  [0.925 0.856 0.742 0.557 0.298 0.081] maxerr 0.126
win 0.1-4 k r_m=1 A k/(k+kappa)              log      [0.657 0.516 0.376 0.237 0.109 0.027] maxerr 0.350
win 0.1-4 k r_m=1 B exp(-kappa/k)            uniform  [0.915 0.839 0.713 0.508 0.218 0.012] maxerr 0.077 PASS
win 0.1-4 k r_m=1 B exp(-kappa/k)            log      [0.618 0.477 0.337 0.198 0.071 0.003] maxerr 0.389
win 0.1-4 k r_m=1 C 1-exp(-k/kappa)          uniform  [0.943 0.887 0.793 0.635 0.369 0.09 ] maxerr 0.204
win 0.1-4 k r_m=1 C 1-exp(-k/kappa)          log      [0.696 0.555 0.415 0.274 0.133 0.03 ] maxerr 0.311
win 0.1-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   uniform  [0.946 0.891 0.8   0.646 0.387 0.013] maxerr 0.215
win 0.1-4 k r_m=2 wall sqrt(1-3kap^2/4k^2)   log      [0.693 0.552 0.411 0.271 0.13  0.003] maxerr 0.314
win 0.1-4 k r_m=2 A k/(k+kappa)              uniform  [0.941 0.883 0.787 0.628 0.389 0.136] maxerr 0.199
win 0.1-4 k r_m=2 A k/(k+kappa)              log      [0.703 0.563 0.423 0.283 0.149 0.047] maxerr 0.304
win 0.1-4 k r_m=2 B exp(-kappa/k)            uniform  [0.932 0.869 0.762 0.587 0.32  0.051] maxerr 0.156
win 0.1-4 k r_m=2 B exp(-kappa/k)            log      [0.665 0.524 0.383 0.244 0.11  0.015] maxerr 0.343
win 0.1-4 k r_m=2 C 1-exp(-k/kappa)          uniform  [0.956 0.909 0.83  0.697 0.473 0.159] maxerr 0.283
win 0.1-4 k r_m=2 C 1-exp(-k/kappa)          log      [0.742 0.602 0.461 0.321 0.18  0.054] maxerr 0.264
```

```
# SCRIPT: PBH-NU-LONE-PASS  the one rate-ratio form that passed (1 of 32) tested on Test B, which it never saw
```

```
# Kill condition (written first): dies if |G_pred - 0.317| > 0.08 at n=6, eps_dec=2.
import numpy as np
eps=np.linspace(0.1,4,6001); kr=1.0
f=lambda kap,k:np.exp(-kap/k)
for n,ed,meas in [(3,2.0,0.431),(6,2.0,0.317)]:
    pred=np.mean(f((ed/eps)**n,kr*eps)); print('n=%d eps_dec=%.0f  pred %.3f  measured %.3f  diff %.3f  %s'%(n
     ,ed,pred,meas,pred-meas,'dies' if abs(pred-meas)>0.08 else 'survives'))
```

```
n=3 eps_dec=2  pred 0.508  measured 0.431  diff 0.077  survives
n=6 eps_dec=2  pred 0.513  measured 0.317  diff 0.196  dies
```

B.12 Diffusive escape

```
# SCRIPT: K6-DIFFUSIVE-LEADER-G-CURVE
# STATUS DISCLOSURE: this model was written AFTER the Table II G column had been read (the log-log slope of G
     /(1-G) ~ -1.6 suggested the
# exponent); the derivation below is independent of the data except for the choice of decision time eps* and
     of the interpolation rule.
# Derivation (all from Steve's Eq.7,10 and the paper's gap-clock reading, Sec.6/8): units r_m = 1, comoving,
     conformal time tau = eps.
#   kappa = H (e_dec/e)^3  ->  scattering rate a*kappa = sigma(tau) = e_dec^3 / tau^4          [Steve Eq.10 +
     gradual-decoupling law]
#   neutrinos (speed 1) diffuse with D = 1/(3 sigma) while sigma*L >> 1, so the NEUTRINO channel's decay rate
     (its principal Metzler
#   eigenvalue, K1-H2 above) is gamma ~ D (pi/L)^2 and its displacement by tau is  l_diff^2 = c * Int D dtau =
      c * tau^5/(15 e_dec^3).
#   Free-streaming (instantaneous decoupling) displaces by l_ball = tau.  Series (resistance) interpolation 1/
     l = 1/l_diff + 1/l_ball and
#   G := l/l_ball = 1/(1 + l_ball/l_diff) = 1/(1 + sqrt(15/c) (e_dec/e*)^(3/2)).        [c = 1: l^2 = Int D
     dtau ; c = 2: 1-D rms ; c = 6: 3-D rms]
# PRE-STATED PASS (this script): max|G_pred - G_pub| <= 0.10 with eps* = 4.0 (Steve's PF apparent-horizon time
     ) and c = 1. Because c is a convention
#   the other c are reported and the one-parameter best fit is reported so that the size of the freedom is
     visible.
# KILL for the exponent claim: a free-exponent fit must give n = 3/2 within its own 2-sigma bootstrap/fit
     uncertainty, else the diffusion scaling is dead.
# OPPOSITE: ballistic-limited regime reading G = 1 - l_diff/l_ball clipped, and G = exp(-l_ball/l_diff).
import numpy as np
from scipy.optimize import least_squares
ed=np.array([.25,.5,1,2,4,8]); G=np.array([.959,.867,.683,.431,.190,.063])
def V1(e,es=4.0,c=1.0): return 1/(1+np.sqrt(15/c)*(e/es)**1.5)
print('published   ',G)
for es in (3.5,4.0,4.5):
    for c in (1.0,2.0,6.0):
        g=V1(ed,es,c); print('V1 e*=%.1f c=%.0f'%(es,c),np.round(g,3),'max|err| %.3f'%np.max(np.abs(g-G)))
# one-parameter fits
r=least_squares(lambda p:1/(1+p[0]*(ed/4.0)**1.5)-G,[3.0]); A=r.x[0]; print('1-param (exponent 3/2, e*=4):
     A_fit=%.3f  (c-equivalent %.2f)  max|err| %.3f'%(A,15/A**2,np.max(np.abs(r.fun))))
r=least_squares(lambda p:1/(1+np.sqrt(15)*(ed/p[0])**1.5)-G,[4.0]); print('1-param (c=1): e*_fit=%.3f max|err|
      %.3f'%(r.x[0],np.max(np.abs(r.fun))))
# free exponent with covariance
def res(p): return 1/(1+p[0]*(ed/4.0)**p[1])-G
r=least_squares(res,[3.0,1.5]); J=r.jac; s2=np.sum(r.fun**2)/(len(ed)-2); cov=np.linalg.inv(J.T@J)*s2
print('free exponent fit: A=%.3f+-%.3f  n=%.3f+-%.3f  max|err| %.3f'%(r.x[0],np.sqrt(cov[0,0]),r.x[1],np.sqrt(
     cov[1,1]),np.max(np.abs(r.fun))))
print('KILL (exponent): n=1.5 is %.1f sd from fitted n'%(abs(r.x[1]-1.5)/np.sqrt(cov[1,1])))
# opposites
l=np.sqrt(1/15)*(ed/4.0)**-1.5*0+ (4.0**2.5)/np.sqrt(15*ed**3)   # l_diff with c=1, eps*=4
lb=4.0
print('OPP1 G=1-l_diff/l_ball clipped ',np.round(np.clip(1-l/lb,0,1),3),'max|err| %.3f'%np.max(np.abs(np.clip
     (1-l/lb,0,1)-G)))
print('OPP2 G=exp(-l_ball/l_diff)     ',np.round(np.exp(-lb/l),3),'max|err| %.3f'%np.max(np.abs(np.exp(-lb/l)-
     G)))
print('OPP3 quadrature G=(1+15 x^3)^-1/2',np.round((1+15*(ed/4.0)**3)**-.5,3),'max|err| %.3f'%np.max(np.abs
     ((1+15*(ed/4.0)**3)**-.5-G)))
```

```
# predictions (new, untested)
for es,tag in ((4.0,'Gaussian'),(8.0,'alpha=0.3 (decided at eps~7-9, Steve Sec.III B)')):
    print('PREDICT %-48s'%tag,{e:round(float(V1(e,es,1.0)),3) for e in (0.1,1,2,4,8,16)})
# Rnu check: odds linear in R? G is defined as ratio so the model predicts the R_nu-independence of G
print('R_nu enters only through the overall (rate) scale -> G(e_dec) curve independent of R_nu (UNTESTED;
     Steve ran gradual decoupling only at R=0.409)')
```

```
published    [0.959 0.867 0.683 0.431 0.19  0.063]
V1 e*=3.5 c=1 [0.931 0.827 0.628 0.374 0.174 0.07 ] max|err| 0.057
V1 e*=3.5 c=2 [0.95  0.871 0.705 0.458 0.23  0.096] max|err| 0.040
V1 e*=3.5 c=6 [0.971 0.921 0.805 0.594 0.341 0.155] max|err| 0.163
V1 e*=4.0 c=1 [0.943 0.854 0.674 0.422 0.205 0.084] max|err| 0.021
V1 e*=4.0 c=2 [0.959 0.892 0.745 0.508 0.267 0.114] max|err| 0.077
V1 e*=4.0 c=6 [0.976 0.935 0.835 0.641 0.387 0.183] max|err| 0.210
V1 e*=4.5 c=1 [0.952 0.875 0.711 0.466 0.236 0.098] max|err| 0.046
V1 e*=4.5 c=2 [0.965 0.908 0.777 0.552 0.303 0.133] max|err| 0.121
V1 e*=4.5 c=6 [0.98  0.945 0.858 0.681 0.43  0.211] max|err| 0.250
1-param (exponent 3/2, e*=4): A_fit=3.811  (c-equivalent 1.03)  max|err| 0.022
1-param (c=1): e*_fit=4.043 max|err| 0.022
free exponent fit: A=4.204+-0.151  n=1.610+-0.032  max|err| 0.010
KILL (exponent): n=1.5 is 3.4 sd from fitted n
OPP1 G=1-l_diff/l_ball clipped  [0.    0.    0.    0.27  0.742 0.909] max|err| 0.959
OPP2 G=exp(-l_ball/l_diff)      [0.941 0.843 0.616 0.254 0.021 0.   ] max|err| 0.177
OPP3 quadrature G=(1+15 x^3)^-1/2 [0.998 0.986 0.9   0.59  0.25  0.091] max|err| 0.217
PREDICT Gaussian                                        {0.1: 0.985, 1: 0.674, 2: 0.422, 4: 0.205, 8: 0.084,
     16: 0.031}
PREDICT alpha=0.3 (decided at eps~7-9, Steve Sec.III B)  {0.1: 0.995, 1: 0.854, 2: 0.674, 4: 0.422, 8: 0.205,
     16: 0.084}
R_nu enters only through the overall (rate) scale -> G(e_dec) curve independent of R_nu (UNTESTED; Steve ran
     gradual decoupling only at R=0.409)
```

B.13 Tests C and D and the controls: predictions, search script and run logs

```
# SCRIPT: PREREGISTERED_NEXT  (written 2026-10-31T17:09:37Z, before any run below)
Follow-up to "When the Neutrinos Leave" v2.  Frozen predictor: kept-sector memory model, two-epoch weight w
     =0.890 (fitted on n=3 only),
values taken unchanged from paper/code/main/memory_curve.json (file written 2026-10-30 13:47 CDT, never
     refitted).

TEST C (fresh history, out of sample): Gaussian profile (alpha=1), rate kappa/H=(eps_dec/eps)^6, eps_dec=1.
  Memory model predicts G = 0.6625.  History-blind rule (Steve's n=3 measured G at the same eps_dec, Table II)
      = 0.683.
  Kill: memory model dies if |G_measured - 0.6625| > 0.08.  History-blind dies if |G - 0.683| > 0.08.
  (The two are only 0.02 apart here, so this test checks the model's survival, not discrimination.)
TEST D (fresh history, discriminating): Gaussian, n=6, eps_dec=3.
  Memory model predicts G = 0.1176.  History-blind (Steve n=3, eps_dec interpolated at 3: between 0.431 (2)
     and 0.190 (4)) ~ 0.30.
  Kill: memory model dies if |G - 0.1176| > 0.08.  History-blind dies if |G - 0.30| > 0.08.

CONTROLS on Test B (n=6, eps_dec=2, measured G=0.317+-0.004 at N=400, K=32, kapcap=5e5):
  B-res: N=600, K=48.  B-cap: kapcap=5e6.  Pass if each shifts delta_c by less than 0.0005 (i.e. G by < 0.017)
     .
  If a control fails, Test B's G is reported as resolution/cap dependent and the 0.071 pass is suspended until
      resolved.
```

```
# SCRIPT: TRISECT3  three-valued two-point bracket search for delta_c (collapse / disperse / undecided-or-
     error)
# Usage: trisect3.py TAG qshape eps_dec nexp lo hi tol eps_max kapcap N K
# Outcome of one run: 'C' (collapse = status BH), 'D' (disperse = status dispersed), 'U' (anything else:
     undecided at eps_max, crash).
# An 'U' point NEVER moves the bracket. It is retried once with eps_max*1.5; if still 'U' it is recorded in '
     undecided' and reported.
import sys, json, time, os
from multiprocessing import Pool
```

```
sys.path.insert(0, os.environ['SIMDIR']); sys.path.insert(0, os.path.dirname(os.path.abspath(__file__)))

def classify(status):
    return 'C' if status == 'BH' else ('D' if status == 'dispersed' else 'U')

def one(a):
    from run_nuc_n3 import run
    d, q, ed, n, em, kc, N, K = a
    t1 = time.time()
    try:
        o = run(d, fnu=0.409, N=N, K=K, eps0=0.1, eps_dec=ed, eps_max=em, qshape=q, nexp=n, kapcap=kc)
        info = dict(d=d, status=o['status'], eps=float(o['eps']), eps_max=em, rawmax=float(o['rawmax']),
     kapcap=kc,
                    capsteps=int(o['capsteps']), steps=int(o['steps']), eps_cap_last=float(o['eps_cap_last']),
      sec=time.time()-t1)
    except Exception as ex:
        info = dict(d=d, status='error:' + type(ex).__name__ + ':' + str(ex)[:80], eps=float('nan'), eps_max=
     em, rawmax=float('nan'),
                    kapcap=kc, capsteps=-1, steps=-1, eps_cap_last=float('nan'), sec=time.time()-t1)
    info['out'] = classify(info['status'])
    return info

def fmt(tag, i):
    return (f"{tag} delta={i['d']:.5f}: {i['out']} [{i['status']}] eps={i['eps']:.2f} eps_max={i['eps_max']} "
            f"rawmax(kappa/H)={i['rawmax']:.3e} cap={i['kapcap']:.1e} capped_steps={i['capsteps']}/{i['steps']
     } "
            f"cap_released_at_eps~{i['eps_cap_last']:.3f} ({i['sec']:.0f}s)")

def search(tag, q, ed, n, lo, hi, tol, em, kc, N, K, mapper, log_path=None, retry_factor=1.5):
    log = []; undecided = []; nonmono = False; t0 = time.time()
    while hi - lo > tol:
        a = lo + (hi - lo) / 3; b = lo + 2 * (hi - lo) / 3
        res = mapper(one, [(a, q, ed, n, em, kc, N, K), (b, q, ed, n, em, kc, N, K)])
        for i in res: print(fmt(tag, i), flush=True); log.append(i)
        # retry each undecided/error point ONCE with raised eps_max (the other core is used for the second one
     )
        bad = [k for k, i in enumerate(res) if i['out'] == 'U']
        if bad:
            rr = mapper(one, [(res[k]['d'], q, ed, n, em * retry_factor, kc, N, K) for k in bad])
            for k, i in zip(bad, rr):
                i['retry'] = True; print('RETRY ' + fmt(tag, i), flush=True); log.append(i)
                if i['out'] == 'U':
                    undecided.append((i['d'], i['status'], i['eps']))
                res[k] = i
        ia, ib = res
        oa, ob = ia['out'], ib['out']
        if oa == 'U' and ob == 'U':
            print(f'{tag} STOP: both points undecided/error after retry; bracket [{lo:.5f},{hi:.5f}] unchanged
     ', flush=True); break
        if oa == 'D' and ob == 'C':     # ordinary case, delta_c between a and b
            lo, hi = a, b
        elif oa == 'C' and ob == 'D':
            print(f'{tag} STOP: non-monotonic outcomes (collapse at a, disperse at b); bracket unchanged',
     flush=True); nonmono = True; break
        elif oa == 'C':                 # a collapses (b is C or U): threshold below a
            hi = a
        elif ob == 'D':                 # b disperses (a is D or U): threshold above b
            lo = b
        elif oa == 'D':                 # a disperses, b undecided: threshold above a
            lo = a
        elif ob == 'C':                 # b collapses, a undecided: threshold below b
            hi = b
        if log_path:
            json.dump(dict(tag=tag, lo=lo, hi=hi, log=log, undecided=undecided), open(log_path, 'w'))
        print(f'{tag} bracket now [{lo:.5f},{hi:.5f}]', flush=True)
    res = dict(tag=tag, qshape=q, eps_dec=ed, nexp=n, kapcap=kc, N=N, K=K, lo=lo, hi=hi, dc=0.5 * (lo + hi), w
     =0.5 * (hi - lo),
               minutes=(time.time() - t0) / 60, undecided=undecided, nonmonotonic=nonmono, log=log)
```

```
    print(f'RESULT {tag}: delta_c = {0.5*(lo+hi):.5f} +- {0.5*(hi-lo):.5f} bracket [{lo:.5f},{hi:.5f}] ({res["
     minutes"]:.1f} min) undecided_points={undecided}', flush=True)
    return res

if __name__ == '__main__':
    tag = sys.argv[1]; q = float(sys.argv[2]); ed = float(sys.argv[3]); n = float(sys.argv[4])
    lo, hi, tol = float(sys.argv[5]), float(sys.argv[6]), float(sys.argv[7]); em = float(sys.argv[8])
    kc = float(sys.argv[9]); N = int(sys.argv[10]); K = int(sys.argv[11])
    with Pool(2) as p:
        res = search(tag, q, ed, n, lo, hi, tol, em, kc, N, K, lambda f, args: p.map(f, args), log_path=f'
     state_{tag}.json')
    json.dump(res, open(f'res_{tag}.json', 'w'))
```

```
Bcap delta=0.50707: D [dispersed] eps=6.25 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250337/5968022 cap_released_at_eps~0.153 (4450s)
Bcap delta=0.50773: C [BH] eps=6.33 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250338/5968766 cap_released_at_eps~0.153 (4465s)
Bcap bracket now [0.50707,0.50773]
```

```
Bcap2 delta=0.50729: D [dispersed] eps=6.54 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250337/5970579 cap_released_at_eps~0.153 (4206s)
Bcap2 delta=0.50751: C [BH] eps=7.29 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+06 capped_steps
     =4250338/5977391 cap_released_at_eps~0.153 (4297s)
Bcap2 bracket now [0.50729,0.50751]
RESULT Bcap2: delta_c = 0.50740 +- 0.00011 bracket [0.50729,0.50751] (71.6 min) undecided_points=[]
```

```
Bres delta=0.50723: D [dispersed] eps=6.37 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809052/1092016 cap_released_at_eps~0.224 (1680s)
Bres delta=0.50757: C [BH] eps=7.32 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809053/1111131 cap_released_at_eps~0.224 (1708s)
Bres bracket now [0.50723,0.50757]
RESULT Bres: delta_c = 0.50740 +- 0.00017 bracket [0.50723,0.50757] (28.5 min) undecided_points=[]
```

```
Bres2 delta=0.50734: D [dispersed] eps=6.53 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809052/1095261 cap_released_at_eps~0.224 (1711s)
Bres2 delta=0.50746: D [dispersed] eps=6.81 eps_max=14.0 rawmax(kappa/H)=6.400e+07 cap=5.0e+05 capped_steps
     =809053/1100808 cap_released_at_eps~0.224 (1712s)
Bres2 bracket now [0.50746,0.50757]
RESULT Bres2: delta_c = 0.50751 +- 0.00006 bracket [0.50746,0.50757] (28.5 min) undecided_points=[]
```

```
C delta=0.50793: D [dispersed] eps=4.77 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/321640 cap_released_at_eps~0.112 (231s)
C delta=0.51812: C [BH] eps=5.91 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps=115550/331789
      cap_released_at_eps~0.112 (240s)
C bracket now [0.50793,0.51812]
C delta=0.51133: D [dispersed] eps=5.00 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/323700 cap_released_at_eps~0.112 (240s)
C delta=0.51472: D [dispersed] eps=5.42 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/327372 cap_released_at_eps~0.112 (246s)
C bracket now [0.51472,0.51812]
C delta=0.51586: D [dispersed] eps=5.68 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/329712 cap_released_at_eps~0.112 (235s)
C delta=0.51699: D [dispersed] eps=6.23 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/334620 cap_released_at_eps~0.112 (239s)
C bracket now [0.51699,0.51812]
C delta=0.51737: D [dispersed] eps=6.81 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps
     =115550/339823 cap_released_at_eps~0.112 (243s)
C delta=0.51775: C [BH] eps=6.32 eps_max=14.0 rawmax(kappa/H)=1.000e+06 cap=5.0e+05 capped_steps=115550/335493
      cap_released_at_eps~0.112 (242s)
C bracket now [0.51737,0.51775]
RESULT C: delta_c = 0.51756 +- 0.00019 bracket [0.51737,0.51775] (16.1 min) undecided_points=[]
```

```
D delta=0.50793: C [BH] eps=4.60 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215105/1415099 cap_released_at_eps~0.337 (1047s)
```

```
D delta=0.51812: C [BH] eps=3.86 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215124/1408481 cap_released_at_eps~0.337 (1039s)
D bracket now [0.49773,0.50793]
D delta=0.50113: D [dispersed] eps=5.74 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215092/1425242 cap_released_at_eps~0.337 (1049s)
D delta=0.50453: C [BH] eps=5.23 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215098/1420654 cap_released_at_eps~0.337 (1054s)
D bracket now [0.50113,0.50453]
D delta=0.50226: D [dispersed] eps=6.42 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215094/1431257 cap_released_at_eps~0.337 (1062s)
D delta=0.50339: C [BH] eps=5.70 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215096/1424836 cap_released_at_eps~0.337 (1069s)
D bracket now [0.50226,0.50339]
D delta=0.50264: C [BH] eps=6.65 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215095/1433339 cap_released_at_eps~0.337 (1064s)
D delta=0.50302: C [BH] eps=5.99 eps_max=14.0 rawmax(kappa/H)=7.290e+08 cap=5.0e+05 capped_steps
     =1215095/1427441 cap_released_at_eps~0.337 (1068s)
D bracket now [0.50226,0.50264]
RESULT D: delta_c = 0.50245 +- 0.00019 bracket [0.50226,0.50264] (70.6 min) undecided_points=[]
```

B.14 The matched-pair witness

```
# SCRIPT: PREREGISTERED_WITNESS (preregistration text, not code)
Written: 2026-10-31T17:12:31Z  (UTC, before any production run; no witness numbers computed yet)

MODEL (identical to paper/code/main/memory_curve.py)
 dv/dtau = -(i k M + sigma(tau)(I-P0)) v, orthonormal Legendre moments v_l, l=0..LM,
 M_{l,l-1}=M_{l-1,l}=l/sqrt(4l^2-1), sigma = eps^n / tau^(n+1), n in {3,6}.
 k-weights: 36-node Gauss-Legendre on [0.25,9], weight k^2 exp(-k^2/2), normalised.
 eps_dec fixed at 0.5 (inside the paper's scanned range) so all plants sit before tau=1.7.

PAIR
 base b at tau_s: v0=1, v1=0.2, v2=0.1, rest 0. Pair: b +/- delta e_L, delta=0.05,
 hidden plant L=3 (main) and L=4; opposite check L=1, L=2 (retained plant, hidden matched).
 Both members scaled by the SAME constant to unit norm (keeps the match; needed for the bound).
 Positivity: f(mu)=sum v_l sqrt((2l+1)/2) P_l(mu) >= 0 on a 2001-point mu grid, both members.
 tau_s / eps in {0.1, 0.25, 0.5, 1, 2}  ->  tau_s in {0.05, 0.125, 0.25, 0.5, 1.0}.
 Readout epochs tau = 1.7, 4.0, late 6.0.

READOUTS (difference = member+ minus member-, exact by linearity = 2 delta U(t,s) e_L)
 R_ret  = sqrt(sum_k w ||Dv_{0,1,2}||^2) / sqrt(sum_k w ||b_{0,1,2}||^2)   (PRIMARY)
 R_A    = |A_w+ - A_w-| / mean(A_w+,A_w-),  A_w = sum_k w |v0|^2       (k-weighted monopole)
 R_F    = same with |v1|^2                                              (k-weighted flux)
 WITNESS ON at an epoch  <=>  R_ret > 1e-3 at that epoch.

DERIVED BEFORE RUNNING (block elimination, a=v0, h=(v1,...)):
 In the hidden block the collision term is sigma*I (scalar), so EXACTLY
   U_Q(t,r) = exp(-(Sig(t)-Sig(r))) exp(-i k (t-r) M_QQ),  Sig(t) = -eps^n/(n t^n).
 a' = S(t) - k^2 int_s^t K(t,r) a(r) dr,  K = M01^2 [U_Q(t,r)]_{11},
 S(t) = -i k M01 [U_Q(t,s) h0]_1.  The pair has Da(s)=0, Dh0 = 2 delta e_L, so the
 difference obeys the same Volterra equation driven ONLY by S[Dh0]: no source, no witness.

PREDICTION (stated before running)
 Two channels carry a hidden difference into the retained moments:
  (i) remnant: the hidden plant survives decoupling with factor exp(-(eps/tau_s)^n / n)
      (int_s^inf sigma converges because collisions die off) and then free-streams: O(delta*remnant);
  (ii) leakage during coupling through L damped rungs: ~ delta*(k/sigma(tau_s))^L.
 Expected witness strength ~ delta*max(remnant, leakage), so with delta=0.05:
   tau_s/eps=0.1 : OFF for n=3 and n=6 (remnant e^-333 / e^-1e6, leakage <1e-11)
   tau_s/eps=0.25: OFF both (n=3 remnant 7e-10, leakage ~2e-7)
   tau_s/eps=0.5 : n=3 ON (remnant e^-2.67=0.07), n=6 OFF or marginal (remnant 2e-5)
   tau_s/eps=1,2 : ON both, O(delta)
  n=6 suppresses more than n=3 at the same tau_s/eps for tau_s<eps, less (closer to 1) for tau_s>=eps.
  Opposite check: deep in coupling L=1 > L=2 > L=3 > L=4 (one factor k/sigma per rung);
  after decoupling all four O(delta) and comparable.
```

```
KILL / SUCCESS CONDITIONS
 K1 (witness dead): R_ret <= 1e-3 at both tau=1.7 and 4.0 for every plant with tau_s >= eps
     -> the retained (density/flux/stress) closure suffices; witness killed in this model.
 K2 (not converged): for any ON case, LM 60->120 or rtol 1e-9->1e-11 (atol 1e-12->1e-14)
     changes R_ret, R_A or R_F by >= 1% -> that entry is not reported as a pass.
 K3 (invalid pair): positivity fails for either member -> pair discarded.
 K4 (prediction killed): any tau_s/eps<=0.25 entry ON, or any tau_s/eps>=1 entry OFF.
 K5 (elimination claim killed): Da computed from the Volterra equation driven by the
     independently computed source S (closed-form U_Q, no full-system solve) differs from the
     full-system Da by > 1% of max|Da|, or fails to converge with grid refinement.
 K6 (speed bound violated): max over all trajectories/times of (A')^2/(4 (k^2/3) A(1-A)) > 1+1e-6.
 SUCCESS: K1-K6 all not triggered.
```

```
# SCRIPT: PBH-NU-WITNESS-RUN  matched retained-state witness in the kept-sector model (same object as
     memory_curve.py)
import os, sys, json, time
os.environ['OMP_NUM_THREADS']='1'
import numpy as np, scipy.sparse as sp
from scipy.integrate import solve_ivp
from scipy.special import roots_legendre, eval_legendre
kn,kw=roots_legendre(36); kmin,kmax=0.25,9.0; ks=0.5*(kmax-kmin)*(kn+1)+kmin; wk=0.5*(kmax-kmin)*kw
wE=wk*ks**2*np.exp(-ks**2/2); wE/=wE.sum()
EPS=0.5; DELTA=0.05; TEP=[1.7,4.0,6.0]
def Mmat(LM):
    d=np.array([l/np.sqrt(4.0*l*l-1) for l in range(1,LM+1)])
    return sp.diags([d,d],[-1,1],format='csr')
def evolve(k,n,ts,LM,rtol,atol,cols):
    N=LM+1; M=Mmat(LM); Q=sp.diags(np.r_[0.0,np.ones(LM)],format='csr'); I=sp.identity(len(cols),format='csr')
    def A(t): return (-1j*k*M - (EPS**n/t**(n+1))*Q)
    def rhs(t,y): return (A(t)@y.reshape(len(cols),N).T).T.ravel()
    def jac(t,y): return sp.kron(I,A(t),format='csc')
    Y0=np.zeros((len(cols),N),complex)
    for j,c in enumerate(cols): Y0[j,:len(c)]=c
    tg=np.unique(np.r_[np.geomspace(ts,6.0,500),TEP])
    s=solve_ivp(rhs,(ts,6.0),Y0.ravel(),method='BDF',t_eval=tg,jac=jac,rtol=rtol,atol=atol)
    assert s.success, s.message
    return s.t, s.y.reshape(len(cols),N,-1)   # [col, l, t]
def positivity(v):
    mu=np.linspace(-1,1,2001)
    return float(min(sum(v[l]*np.sqrt((2*l+1)/2)*eval_legendre(l,mu) for l in range(len(v)))))
def run(n,ratio,L,LM=60,rtol=1e-9,atol=1e-12):
    ts=ratio*EPS; b=np.array([1.0,0.2,0.1,0,0]); e=np.zeros(5); e[L]=1
    c=1/np.sqrt(b@b+DELTA**2)
    pos=(positivity(c*(b+DELTA*e)),positivity(c*(b-DELTA*e)))
    rows={tt:dict(dret=0.0,bret=0.0,Ap=0.0,Am=0.0,Fp=0.0,Fm=0.0) for tt in TEP}
    worst=0.0; worstN=0.0
    for k,w in zip(ks,wE):
        t,Y=evolve(k,n,ts,LM,rtol,atol,[c*b,c*DELTA*e])
        U_b,U_d=Y[0],Y[1]
        for sgn in (1,-1):
            v=U_b+sgn*U_d; A=np.abs(v[0])**2; Ap=2*np.real(np.conj(v[0])*(-1j*k*(1/np.sqrt(3))*v[1]))
            Nrm=np.sum(np.abs(v)**2,axis=0)
            den=4*(k*k/3)*A*(1-A); ok=den>1e-300
            worst=max(worst,float(np.max(Ap[ok]**2/den[ok])))
            denN=4*(k*k/3)*A*(Nrm-A); ok=denN>1e-300
            worstN=max(worstN,float(np.max(Ap[ok]**2/denN[ok])))
        for tt in TEP:
            i=np.argmin(abs(t-tt)); r=rows[tt]
            r['dret']+=w*np.sum(np.abs(2*U_d[:3,i])**2); r['bret']+=w*np.sum(np.abs(U_b[:3,i])**2)
            vp=U_b[:,i]+U_d[:,i]; vm=U_b[:,i]-U_d[:,i]
            r['Ap']+=w*abs(vp[0])**2; r['Am']+=w*abs(vm[0])**2; r['Fp']+=w*abs(vp[1])**2; r['Fm']+=w*abs(vm
     [1])**2
    res={}
    for tt in TEP:
        r=rows[tt]
        res[str(tt)]=dict(R_ret=float(np.sqrt(r['dret']/r['bret'])),
            R_A=float(abs(r['Ap']-r['Am'])/(0.5*(r['Ap']+r['Am']))),
            R_F=float(abs(r['Fp']-r['Fm'])/(0.5*(r['Fp']+r['Fm']))))
```

```
    return dict(n=n,ratio=ratio,tau_s=ts,L=L,LM=LM,rtol=rtol,pos_min=pos,bound_worst=worst,bound_worst_N=
     worstN,res=res)
if __name__=='__main__':
    mode=sys.argv[1]; out=[]
    if mode=='prod':
        jobs=[(n,r,L) for n in (3,6) for r in (0.1,0.25,0.5,1,2) for L in (3,4,1,2)]
        kw_={}
    else:
        jobs=[(n,r,L) for n in (3,6) for r in (0.5,1,2) for L in (3,4)]
        kw_=dict(LM=120) if mode=='convLM' else dict(rtol=1e-11,atol=1e-14)
    for j in jobs:
        t0=time.time(); d=run(*j,**kw_); out.append(d)
        print(j,'%.1fs'%(time.time()-t0),' '.join('%s:%.3e/%.3e/%.3e'%(tt,v['R_ret'],v['R_A'],v['R_F']) for tt
     ,v in d['res'].items()),
              'pos %.3f %.3f bound %.6f %.6f'%(*d['pos_min'],d['bound_worst'],d['bound_worst_N']),flush=True)
    json.dump(out,open('witness_%s.json'%mode,'w'),indent=1)
```

```
# SCRIPT: PBH-NU-WITNESS-KERNEL  exact block elimination: hidden source S = -ik B U_Q(t,s) Dh0 drives the
     whole retained difference
import os; os.environ['OMP_NUM_THREADS']='1'
import numpy as np, json, scipy.sparse as sp
from scipy.integrate import solve_ivp
EPS=0.5; DELTA=0.05; LM=60; N=LM+1
M=np.zeros((N,N))
for l in range(1,N): M[l,l-1]=M[l-1,l]=l/np.sqrt(4.0*l*l-1)
MQQ=M[1:,1:]; lam,V=np.linalg.eigh(MQQ); M01=M[0,1]
Sig=lambda t,n: -EPS**n/(n*t**n)            # Sig' = sigma
def full_Da(k,n,ts,L,tg,c):
    Q=np.diag(np.r_[0.0,np.ones(LM)])
    A=lambda t: -1j*k*M-(EPS**n/t**(n+1))*Q
    y0=np.zeros(N,complex); y0[L]=2*DELTA*c
    s=solve_ivp(lambda t,y:A(t)@y,(ts,tg[-1]),y0,method='BDF',jac=lambda t,y:A(t),t_eval=tg,rtol=1e-12,atol=1e
     -15)
    return s.y[0]
def memory_Da(k,n,ts,L,tg,c):
    # closed-form hidden propagator, no full-system solve
    h=tg[1]-tg[0]; Nt=len(tg); Sg=Sig(tg,n)
    Dh0=np.zeros(LM,complex); Dh0[L-1]=2*DELTA*c
    coef=V.T@Dh0
    S=-1j*k*M01*np.exp(-(Sg-Sg[0]))*((V[0,:]*coef)[None,:]*np.exp(-1j*k*np.outer(tg-ts,lam))).sum(1)
    g=M01**2*((V[0,:]**2)[None,:]*np.exp(-1j*k*np.outer(np.arange(Nt)*h,lam))).sum(1)   # g(t-r)
    a=np.zeros(Nt,complex); F=np.zeros(Nt,complex); F[0]=S[0]
    for i in range(Nt-1):
        j=np.arange(i+2); Kr=np.exp(-(Sg[i+1]-Sg[j]))*g[i+1-j]
        w=np.full(i+2,h); w[0]=w[-1]=h/2
        Jknown=np.sum((w*Kr*a[j])[:-1])
        # a_{i+1} = a_i + h/2 (F_i + S_{i+1} - k^2 (Jknown + w_last K_ii a_{i+1}))
        a[i+1]=(a[i]+h/2*(F[i]+S[i+1]-k*k*Jknown))/(1+h/2*k*k*w[-1]*Kr[-1])
        F[i+1]=S[i+1]-k*k*(Jknown+w[-1]*Kr[-1]*a[i+1])
    Sint=np.concatenate([[0],np.cumsum(h/2*(S[1:]+S[:-1]))])
    return a,Sint
out=[]
b=np.array([1.0,0.2,0.1]); c=1/np.sqrt(b@b+DELTA**2)
for (n,ratio,L,k) in [(3,1,3,2.0),(3,0.5,3,1.5),(6,1,4,3.0),(6,2,3,1.0)]:
    ts=ratio*EPS; row=dict(n=n,ratio=ratio,L=L,k=k)
    for h in (4e-3,2e-3,1e-3):
        tg=ts+np.arange(int(round((4.0-ts)/h))+1)*h
        ref=full_Da(k,n,ts,L,tg,c); a,Sint=memory_Da(k,n,ts,L,tg,c)
        err=np.max(abs(a-ref))/np.max(abs(ref))
        row['h=%g'%h]=dict(rel_err=float(err),max_Da=float(np.max(abs(ref))),
            src_only_rel=float(np.max(abs(Sint-ref))/np.max(abs(ref))))
        print(n,ratio,L,k,h,'max|Da|=%.3e  memory-vs-full rel err=%.2e  (source integral alone, no kernel: %.2
     e)'%(np.max(abs(ref)),err,row['h=%g'%h]['src_only_rel']),flush=True)
    out.append(row)
json.dump(out,open('witness_kernel.json','w'),indent=1)
```

```
# SCRIPT: PBH-NU-WITNESS-CONV  compare production vs LM=120 and tight-tolerance runs (K2)
```

```
import json
P={(d['n'],d['ratio'],d['L']):d for d in json.load(open('witness_prod.json'))}
worst=0
for mode in ('convLM','convTol'):
    for d in json.load(open('witness_%s.json'%mode)):
        p=P[(d['n'],d['ratio'],d['L'])]; ch=[]
        for tt in ('1.7','4.0','6.0'):
            for q in ('R_ret','R_A','R_F'):
                ch.append(abs(d['res'][tt][q]-p['res'][tt][q])/abs(p['res'][tt][q]))
        worst=max(worst,max(ch)); print(mode,d['n'],d['ratio'],d['L'],'max rel change %.2e'%max(ch),'bound %.6
    f'%d['bound_worst'])
print('WORST',worst)
```


```
[
 {
  "n": 3,
  "ratio": 1,
  "L": 3,
  "k": 2.0,
  "h=0.004": {
   "rel_err": 4.742985300596294e-06,
   "max_Da": 0.04492003160306953,
   "src_only_rel": 1.9909068121545908
  },
  "h=0.002": {
   "rel_err": 1.1856044767257576e-06,
   "max_Da": 0.04492007612793319,
   "src_only_rel": 1.990905398421171
  },
  "h=0.001": {
   "rel_err": 2.9625562593651493e-07,
   "max_Da": 0.044920090604062296,
   "src_only_rel": 1.9909048967372047
  }
 },
 {
  "n": 3,
  "ratio": 0.5,
  "L": 3,
  "k": 1.5,
  "h=0.004": {
   "rel_err": 1.9702037965960922e-06,
   "max_Da": 0.00434791177399913,
   "src_only_rel": 1.2648986553695796
  },
  "h=0.002": {
   "rel_err": 4.932862271426013e-07,
   "max_Da": 0.00434791177399913,
   "src_only_rel": 1.2630671055573819
  },
  "h=0.001": {
   "rel_err": 1.2404572232928826e-07,
   "max_Da": 0.00434791177399913,
   "src_only_rel": 1.2630672812226218
  }
 },
 {
  "n": 6,
  "ratio": 1,
  "L": 4,
  "k": 3.0,
  "h=0.004": {
   "rel_err": 1.1862372761170309e-05,
   "max_Da": 0.049899459055607595,
   "src_only_rel": 2.3445679331226095
  },
  "h=0.002": {
   "rel_err": 2.965509149910457e-06,
```

```
   "max_Da": 0.049899739333988655,
   "src_only_rel": 2.3445545251342654
  },
  "h=0.001": {
   "rel_err": 7.412824292084345e-07,
   "max_Da": 0.049899739333988655,
   "src_only_rel": 2.3445558282676617
  }
 },
 {
  "n": 6,
  "ratio": 2,
  "L": 3,
  "k": 1.0,
  "h=0.004": {
   "rel_err": 7.922658929918443e-07,
   "max_Da": 0.03911120365950552,
   "src_only_rel": 0.1881606736764026
  },
  "h=0.002": {
   "rel_err": 1.9809644902127958e-07,
   "max_Da": 0.03911120365950552,
   "src_only_rel": 0.18816047472879305
  },
  "h=0.001": {
   "rel_err": 4.955392667016778e-08,
   "max_Da": 0.03911120365950552,
   "src_only_rel": 0.18816042499176522
  }
 }
]
```


```
(3, 0.1, 3) 18.4s 1.7:1.445e-14/0.000e+00/0.000e+00 4.0:7.318e-15/0.000e+00/0.000e+00 6.0:6.950e-15/0.000e
    +00/0.000e+00 pos 0.513 0.510 bound 0.788639 1.000001
(3, 0.1, 4) 20.3s 1.7:9.639e-19/0.000e+00/0.000e+00 4.0:4.764e-19/0.000e+00/0.000e+00 6.0:4.547e-19/0.000e
    +00/0.000e+00 pos 0.510 0.501 bound 0.788639 1.000001
(3, 0.1, 1) 18.2s 1.7:4.619e-06/4.227e-11/8.959e-11 4.0:3.230e-06/3.279e-11/3.619e-11 6.0:2.986e-06/3.647e
    -11/3.323e-11 pos 0.516 0.578 bound 0.788639 1.000001
(3, 0.1, 2) 17.8s 1.7:2.415e-10/1.898e-10/4.023e-10 4.0:1.349e-10/1.472e-10/1.625e-10 6.0:1.264e-10/1.638e
    -10/1.492e-10 pos 0.533 0.528 bound 0.788639 1.000001
(3, 0.25, 3) 16.1s 1.7:1.326e-09/2.388e-13/8.056e-13 4.0:6.311e-10/1.919e-13/2.070e-13 6.0:5.913e-10/2.028e
    -13/1.873e-13 pos 0.513 0.510 bound 0.788648 1.000000
(3, 0.25, 4) 15.3s 1.7:2.953e-11/8.962e-12/3.087e-11 4.0:6.182e-11/9.579e-11/6.599e-11 6.0:8.659e-11/1.151e
    -10/6.062e-11 pos 0.510 0.501 bound 0.788648 1.000000
(3, 0.25, 1) 16.6s 1.7:1.924e-04/7.336e-08/1.555e-07 4.0:1.345e-04/5.690e-08/6.280e-08 6.0:1.244e-04/6.329e
    -08/5.767e-08 pos 0.516 0.578 bound 0.788648 1.000000
(3, 0.25, 2) 16.3s 1.7:4.516e-07/3.551e-07/7.523e-07 4.0:2.522e-07/2.754e-07/3.039e-07 6.0:2.364e-07/3.062e
    -07/2.790e-07 pos 0.533 0.528 bound 0.788648 1.000000
(3, 0.5, 3) 15.7s 1.7:4.924e-03/1.185e-04/9.109e-05 4.0:8.557e-03/5.292e-05/2.266e-04 6.0:1.120e-02/3.412e
    -04/3.867e-04 pos 0.513 0.510 bound 0.794143 0.991388
(3, 0.5, 4) 16.6s 1.7:3.480e-03/6.867e-04/4.006e-03 4.0:7.905e-03/1.143e-02/7.323e-03 6.0:1.101e-02/1.203e
    -02/5.594e-03 pos 0.510 0.501 bound 0.794396 0.991416
(3, 0.5, 1) 15.1s 1.7:9.487e-03/5.249e-04/2.917e-04 4.0:1.096e-02/7.344e-04/4.928e-04 6.0:1.163e-02/8.143e
    -04/5.617e-04 pos 0.516 0.578 bound 0.794207 0.994604
(3, 0.5, 2) 16.0s 1.7:5.897e-03/7.879e-03/7.402e-03 4.0:8.996e-03/2.506e-03/8.059e-03 6.0:1.094e-02/1.278e
    -02/1.289e-02 pos 0.533 0.528 bound 0.796846 0.993249
(3, 1, 3) 12.1s 1.7:4.654e-02/7.218e-03/1.065e-03 4.0:8.123e-02/7.397e-03/1.277e-02 6.0:1.052e-01/1.536e
    -02/2.445e-02 pos 0.513 0.510 bound 0.828287 0.891815
(3, 1, 4) 11.4s 1.7:2.895e-02/3.765e-03/2.858e-02 4.0:7.420e-02/1.019e-01/5.072e-02 6.0:1.028e-01/7.643e
    -02/2.026e-02 pos 0.510 0.501 bound 0.828860 0.892318
(3, 1, 1) 10.9s 1.7:7.130e-02/2.942e-02/1.165e-02 4.0:8.961e-02/5.652e-02/3.277e-02 6.0:9.713e-02/5.837e
    -02/3.891e-02 pos 0.516 0.578 bound 0.836234 0.932473
(3, 1, 2) 9.7s 1.7:5.852e-02/5.858e-02/9.228e-02 4.0:8.650e-02/7.613e-02/1.029e-01 6.0:1.038e-01/1.652e
    -01/1.477e-01 pos 0.533 0.528 bound 0.854802 0.906232
(3, 2, 3) 8.5s 1.7:4.719e-02/1.900e-03/6.876e-03 4.0:9.529e-02/2.314e-02/1.613e-02 6.0:1.291e-01/1.467e
    -02/3.407e-02 pos 0.513 0.510 bound 0.834221 0.844598
(3, 2, 4) 7.9s 1.7:1.832e-02/1.597e-03/1.464e-02 4.0:8.500e-02/1.176e-01/5.982e-02 6.0:1.244e-01/9.771e
    -02/2.459e-02 pos 0.510 0.501 bound 0.834124 0.844526
```

```
(3, 2, 1) 8.3s 1.7:9.340e-02/1.884e-02/4.561e-02 4.0:1.096e-01/8.991e-02/4.796e-02 6.0:1.210e-01/9.091e
    -02/5.989e-02 pos 0.516 0.578 bound 0.848836 0.902645
(3, 2, 2) 7.7s 1.7:8.247e-02/3.742e-02/1.390e-01 4.0:1.039e-01/8.610e-02/1.267e-01 6.0:1.292e-01/2.062e
    -01/1.870e-01 pos 0.533 0.528 bound 0.868966 0.860951
(6, 0.1, 3) 17.4s 1.7:1.473e-23/0.000e+00/0.000e+00 4.0:7.797e-24/0.000e+00/0.000e+00 6.0:7.425e-24/0.000e
    +00/0.000e+00 pos 0.513 0.510 bound 0.795402 1.479631
(6, 0.1, 4) 18.4s 1.7:9.800e-31/0.000e+00/0.000e+00 4.0:5.108e-31/0.000e+00/0.000e+00 6.0:4.880e-31/0.000e
    +00/0.000e+00 pos 0.510 0.501 bound 0.795402 1.479636
(6, 0.1, 1) 17.5s 1.7:4.672e-09/0.000e+00/0.000e+00 4.0:3.300e-09/2.080e-16/0.000e+00 6.0:3.075e-09/0.000e
    +00/0.000e+00 pos 0.516 0.578 bound 0.795402 1.479637
(6, 0.1, 2) 18.3s 1.7:2.458e-16/0.000e+00/2.907e-16 4.0:1.414e-16/0.000e+00/1.377e-16 6.0:1.333e-16/1.393e
    -16/0.000e+00 pos 0.533 0.528 bound 0.795402 1.479631
(6, 0.25, 3) 17.0s 1.7:3.384e-15/0.000e+00/0.000e+00 4.0:1.791e-15/0.000e+00/0.000e+00 6.0:1.706e-15/0.000e
    +00/0.000e+00 pos 0.513 0.510 bound 0.795405 1.000002
(6, 0.25, 4) 16.4s 1.7:1.383e-19/0.000e+00/0.000e+00 4.0:7.210e-20/0.000e+00/0.000e+00 6.0:6.890e-20/0.000e
    +00/0.000e+00 pos 0.510 0.501 bound 0.795405 1.000002
(6, 0.25, 1) 17.2s 1.7:2.857e-06/1.570e-11/3.354e-11 4.0:2.018e-06/1.325e-11/1.429e-11 6.0:1.880e-06/1.457e
    -11/1.326e-11 pos 0.516 0.578 bound 0.795405 1.000002
(6, 0.25, 2) 17.6s 1.7:9.203e-11/7.033e-11/1.502e-10 4.0:5.293e-11/5.935e-11/6.400e-11 6.0:4.993e-11/6.526e
    -11/5.941e-11 pos 0.533 0.528 bound 0.795405 1.000002
(6, 0.5, 3) 15.0s 1.7:1.589e-06/6.042e-10/2.030e-09 4.0:2.816e-06/3.803e-09/2.994e-09 6.0:3.699e-06/4.842e
    -09/3.566e-09 pos 0.513 0.510 bound 0.795417 0.999989
(6, 0.5, 4) 15.4s 1.7:1.126e-06/2.047e-07/1.299e-06 4.0:2.609e-06/3.835e-06/2.372e-06 6.0:3.643e-06/3.727e
    -06/1.678e-06 pos 0.510 0.501 bound 0.795417 0.999989
(6, 0.5, 1) 15.2s 1.7:4.181e-04/3.412e-07/7.185e-07 4.0:2.961e-04/2.866e-07/3.083e-07 6.0:2.759e-04/3.142e
    -07/2.859e-07 pos 0.516 0.578 bound 0.795417 0.999992
(6, 0.5, 2) 14.2s 1.7:2.880e-06/4.291e-06/1.372e-06 4.0:2.592e-06/3.594e-07/1.237e-06 6.0:3.051e-06/2.823e
    -06/2.960e-06 pos 0.533 0.528 bound 0.795418 0.999992
(6, 1, 3) 11.7s 1.7:5.345e-02/9.702e-03/8.884e-04 4.0:9.377e-02/1.032e-02/1.629e-02 6.0:1.207e-01/1.853e
    -02/3.100e-02 pos 0.513 0.510 bound 0.835375 0.867593
(6, 1, 4) 11.3s 1.7:3.326e-02/3.456e-03/3.198e-02 4.0:8.570e-02/1.157e-01/5.204e-02 6.0:1.179e-01/7.385e
    -02/1.193e-02 pos 0.510 0.501 bound 0.835639 0.867877
(6, 1, 1) 11.0s 1.7:8.003e-02/3.822e-02/1.457e-02 4.0:1.010e-01/7.239e-02/4.155e-02 6.0:1.094e-01/7.339e
    -02/4.928e-02 pos 0.516 0.578 bound 0.846711 0.917310
(6, 1, 2) 11.3s 1.7:6.703e-02/6.418e-02/1.060e-01 4.0:9.957e-02/1.071e-01/1.271e-01 6.0:1.189e-01/2.012e
    -01/1.769e-01 pos 0.533 0.528 bound 0.866573 0.883317
(6, 2, 3) 8.6s 1.7:4.848e-02/1.999e-03/7.186e-03 4.0:9.826e-02/2.450e-02/1.702e-02 6.0:1.329e-01/1.523e
    -02/3.585e-02 pos 0.513 0.510 bound 0.836364 0.836927
(6, 2, 4) 8.6s 1.7:1.882e-02/1.634e-03/1.490e-02 4.0:8.768e-02/1.221e-01/6.027e-02 6.0:1.282e-01/9.738e
    -02/2.238e-02 pos 0.510 0.501 bound 0.836144 0.836710
(6, 2, 1) 8.5s 1.7:9.572e-02/1.969e-02/4.752e-02 4.0:1.123e-01/9.466e-02/5.035e-02 6.0:1.240e-01/9.520e
    -02/6.286e-02 pos 0.516 0.578 bound 0.852239 0.897805
(6, 2, 2) 8.2s 1.7:8.471e-02/3.832e-02/1.416e-01 4.0:1.070e-01/9.479e-02/1.327e-01 6.0:1.329e-01/2.159e
    -01/1.947e-01 pos 0.533 0.528 bound 0.872473 0.853435
```

```
(3, 0.5, 3) 13.1s 1.7:4.924e-03/1.185e-04/9.109e-05 4.0:8.557e-03/5.292e-05/2.266e-04 6.0:1.120e-02/3.412e
    -04/3.867e-04 pos 0.513 0.510 bound 0.794143 0.991388
(3, 0.5, 4) 14.8s 1.7:3.480e-03/6.867e-04/4.006e-03 4.0:7.905e-03/1.143e-02/7.323e-03 6.0:1.101e-02/1.203e
    -02/5.594e-03 pos 0.510 0.501 bound 0.794396 0.991416
(3, 1, 3) 12.2s 1.7:4.654e-02/7.218e-03/1.065e-03 4.0:8.123e-02/7.397e-03/1.277e-02 6.0:1.052e-01/1.536e
    -02/2.445e-02 pos 0.513 0.510 bound 0.828287 0.891815
(3, 1, 4) 12.1s 1.7:2.895e-02/3.765e-03/2.858e-02 4.0:7.420e-02/1.019e-01/5.072e-02 6.0:1.028e-01/7.643e
    -02/2.026e-02 pos 0.510 0.501 bound 0.828860 0.892318
(3, 2, 3) 9.8s 1.7:4.719e-02/1.900e-03/6.876e-03 4.0:9.529e-02/2.314e-02/1.613e-02 6.0:1.291e-01/1.467e
    -02/3.407e-02 pos 0.513 0.510 bound 0.834221 0.844598
(3, 2, 4) 9.4s 1.7:1.832e-02/1.597e-03/1.464e-02 4.0:8.500e-02/1.176e-01/5.982e-02 6.0:1.244e-01/9.771e
    -02/2.459e-02 pos 0.510 0.501 bound 0.834124 0.844526
(6, 0.5, 3) 16.4s 1.7:1.589e-06/6.042e-10/2.030e-09 4.0:2.816e-06/3.803e-09/2.994e-09 6.0:3.699e-06/4.842e
    -09/3.566e-09 pos 0.513 0.510 bound 0.795417 0.999989
(6, 0.5, 4) 17.0s 1.7:1.126e-06/2.047e-07/1.299e-06 4.0:2.609e-06/3.835e-06/2.372e-06 6.0:3.643e-06/3.727e
    -06/1.678e-06 pos 0.510 0.501 bound 0.795417 0.999989
(6, 1, 3) 11.9s 1.7:5.345e-02/9.702e-03/8.884e-04 4.0:9.377e-02/1.032e-02/1.629e-02 6.0:1.207e-01/1.853e
    -02/3.100e-02 pos 0.513 0.510 bound 0.835375 0.867593
(6, 1, 4) 13.1s 1.7:3.326e-02/3.456e-03/3.198e-02 4.0:8.570e-02/1.157e-01/5.204e-02 6.0:1.179e-01/7.385e
    -02/1.193e-02 pos 0.510 0.501 bound 0.835639 0.867877
(6, 2, 3) 9.3s 1.7:4.848e-02/1.999e-03/7.186e-03 4.0:9.826e-02/2.450e-02/1.702e-02 6.0:1.329e-01/1.523e
    -02/3.585e-02 pos 0.513 0.510 bound 0.836364 0.836927
```

```
(6, 2, 4) 9.1s 1.7:1.882e-02/1.634e-03/1.490e-02 4.0:8.768e-02/1.221e-01/6.027e-02 6.0:1.282e-01/9.738e
    -02/2.238e-02 pos 0.510 0.501 bound 0.836144 0.836710
```

```
(3, 0.5, 3) 30.5s 1.7:4.924e-03/1.185e-04/9.109e-05 4.0:8.557e-03/5.292e-05/2.266e-04 6.0:1.120e-02/3.412e
    -04/3.867e-04 pos 0.513 0.510 bound 0.794143 0.991388
(3, 0.5, 4) 31.6s 1.7:3.480e-03/6.867e-04/4.006e-03 4.0:7.905e-03/1.143e-02/7.323e-03 6.0:1.101e-02/1.203e
    -02/5.594e-03 pos 0.510 0.501 bound 0.794396 0.991416
(3, 1, 3) 19.7s 1.7:4.654e-02/7.218e-03/1.065e-03 4.0:8.123e-02/7.397e-03/1.277e-02 6.0:1.052e-01/1.536e
    -02/2.445e-02 pos 0.513 0.510 bound 0.828287 0.891815
(3, 1, 4) 19.6s 1.7:2.895e-02/3.765e-03/2.858e-02 4.0:7.420e-02/1.019e-01/5.072e-02 6.0:1.028e-01/7.643e
    -02/2.026e-02 pos 0.510 0.501 bound 0.828860 0.892318
(3, 2, 3) 17.9s 1.7:4.719e-02/1.900e-03/6.876e-03 4.0:9.529e-02/2.314e-02/1.613e-02 6.0:1.291e-01/1.467e
    -02/3.407e-02 pos 0.513 0.510 bound 0.834221 0.844598
(3, 2, 4) 18.8s 1.7:1.832e-02/1.597e-03/1.464e-02 4.0:8.500e-02/1.176e-01/5.982e-02 6.0:1.244e-01/9.771e
    -02/2.459e-02 pos 0.510 0.501 bound 0.834124 0.844526
(6, 0.5, 3) 29.7s 1.7:1.589e-06/6.042e-10/2.030e-09 4.0:2.816e-06/3.803e-09/2.994e-09 6.0:3.699e-06/4.842e
    -09/3.566e-09 pos 0.513 0.510 bound 0.795417 0.999989
(6, 0.5, 4) 28.4s 1.7:1.126e-06/2.047e-07/1.299e-06 4.0:2.609e-06/3.834e-06/2.372e-06 6.0:3.643e-06/3.727e
    -06/1.678e-06 pos 0.510 0.501 bound 0.795417 0.999989
(6, 1, 3) 19.3s 1.7:5.345e-02/9.702e-03/8.884e-04 4.0:9.377e-02/1.032e-02/1.629e-02 6.0:1.207e-01/1.853e
    -02/3.100e-02 pos 0.513 0.510 bound 0.835375 0.867593
(6, 1, 4) 21.4s 1.7:3.326e-02/3.456e-03/3.198e-02 4.0:8.570e-02/1.157e-01/5.204e-02 6.0:1.179e-01/7.385e
    -02/1.193e-02 pos 0.510 0.501 bound 0.835639 0.867877
(6, 2, 3) 17.5s 1.7:4.848e-02/1.999e-03/7.186e-03 4.0:9.826e-02/2.450e-02/1.702e-02 6.0:1.329e-01/1.523e
    -02/3.585e-02 pos 0.513 0.510 bound 0.836364 0.836927
(6, 2, 4) 18.5s 1.7:1.882e-02/1.634e-03/1.490e-02 4.0:8.768e-02/1.221e-01/6.027e-02 6.0:1.282e-01/9.738e
    -02/2.238e-02 pos 0.510 0.501 bound 0.836144 0.836710
```

B.15 Figures

```
# SCRIPT: PBH-NU-FIGURES  draws the three figures from the run results and the memory curves
import json, numpy as np, matplotlib
matplotlib.use('Agg'); import matplotlib.pyplot as plt
plt.rcParams.update({'font.family':'DejaVu Serif','font.size':11})
mc=json.load(open('code/main/memory_curve.json'))
ed3=[0.25,0.5,0.75,1,1.5,2,3,4,6,8]; ed6=[0.5,1,1.5,2,3,4]
G3=[mc['3_%g'%e]['two_epoch'] for e in ed3]; G6=[mc['6_%g'%e]['two_epoch'] for e in ed6]
steve_e=[0.25,0.5,1,2,4,8]; steve_G=[0.959,0.867,0.683,0.431,0.190,0.063]
# Fig 1
fig,ax=plt.subplots(figsize=(7.2,4.6))
ax.plot(ed3,G3,'-',color='#1f5fa8',lw=2,label='kept-sector memory, rate $\\propto\\epsilon^{-3}$ (one weight
    fitted here)')
ax.plot(ed6,G6,'--',color='#c0392b',lw=2,label='kept-sector memory, rate $\\propto\\epsilon^{-6}$ (prediction)
    ')
ax.plot(steve_e,steve_G,'o',color='k',ms=7,label='McCormick, Gaussian, rate $\\propto\\epsilon^{-3}$')
ax.plot([4,8],[0.181,0.045],'s',color='#2e8b57',ms=8,label='this work, broad profile $\\alpha=0.3$, rate $\\
    propto\\epsilon^{-3}$')
ax.errorbar([2],[0.317],yerr=[0.004],fmt='D',color='#c0392b',ms=8,label='this work, Gaussian, rate $\\propto\\
    epsilon^{-6}$')
ax.plot([2],[0.431],'x',color='gray',ms=10,mew=2); ax.annotate('if only $\\epsilon_{dec}$ mattered',(2,0.431)
    ,(2.6,0.52),fontsize=9,color='gray',arrowprops=dict(arrowstyle='->',color='gray'))
ax.set_xscale('log'); ax.set_xticks([0.25,0.5,1,2,4,8]); ax.set_xticklabels(['0.25','0.5','1','2','4','8']);
    ax.minorticks_off(); ax.set_xlabel('decoupling time $\\epsilon_{dec}$'); ax.set_ylabel('surviving share
    of the free-streaming shift, $G$')
ax.set_ylim(0,1.02); ax.legend(fontsize=8.2,loc='lower left',frameon=False); ax.set_title('How much of the
    neutrino effect survives gradual decoupling',fontsize=11)
fig.tight_layout(); fig.savefig('figs/fig1_G.png',dpi=220)
# Fig 2: Legendre chain
fig,ax=plt.subplots(figsize=(7.2,2.6)); ax.axis('off')
xs=np.arange(6)*1.2
for i,x in enumerate(xs):
    ax.add_patch(plt.Circle((x,0),0.28,color=['#f4d03f','#e67e22','#c0392b','#8e44ad','#8e44ad','#8e44ad'][i],
    zorder=3))
    ax.text(x,0,'$\\ell=%d$'%i if i<5 else '$\\cdots$',ha='center',va='center',fontsize=10,zorder=4,color='
    white' if i>1 else 'k')
```

```
for i in range(5):
    lw=[6,4,0.8,0.8,0.8][i]; col=['#c0392b','#e67e22','#bbbbbb','#bbbbbb','#bbbbbb'][i]
    ax.plot([xs[i]+0.28,xs[i+1]-0.28],[0,0],color=col,lw=lw,zorder=2)
ax.text((xs[0]+xs[1])/2,0.45,'$k^2\\kappa_1^2/3$',ha='center',fontsize=10,color='#c0392b')
ax.text((xs[1]+xs[2])/2,0.45,'$\\frac{4k^2}{15}(\\kappa_2-\\kappa_1)^2$',ha='center',fontsize=10,color='#
    e67e22')
ax.text((xs[3]+xs[4])/2,0.45,'0',ha='center',fontsize=10,color='gray')
ax.text(xs[0],-0.6,'density\n(kept)',ha='center',fontsize=8.5); ax.text(xs[1],-0.6,'flux\nrate $\\kappa_1$',ha
    ='center',fontsize=8.5); ax.text(xs[2],-0.6,'shear\nrate $\\kappa_2$',ha='center',fontsize=8.5); ax.text(
    xs[4],-0.6,'higher moments\nrate $\\kappa_2$',ha='center',fontsize=8.5)
ax.set_xlim(-0.5,xs[-1]+0.5); ax.set_ylim(-1.0,0.85)
ax.set_title('Where the non-normality of the neutrino collision operator lives',fontsize=11)
fig.tight_layout(); fig.savefig('figs/fig2_chain.png',dpi=220)
# Fig 3: clock set early
fig,ax=plt.subplots(figsize=(7.2,3.6))
ax.bar([0.8,2.8],[0.190,0.063],0.35,color='#555555',label='Gaussian (McCormick): collapses at $\\epsilon\\
    approx4$')
ax.bar([1.2,3.2],[0.181,0.045],0.35,color='#2e8b57',label='broad profile $\\alpha=0.3$ (this work): collapses
    at $\\epsilon\\approx7$-$9$')
ax.plot([0.8,1.2,2.8,3.2],[0.38,0.38,0.17,0.17],'_',ms=0)
ax.hlines([0.42],1.02,1.38,colors='#c0392b',linestyles='dotted',lw=2); ax.hlines([0.205],3.02,3.38,colors='#
    c0392b',linestyles='dotted',lw=2)
ax.text(1.2,0.44,'if the clock were the collapse time',ha='center',fontsize=8.5,color='#c0392b'); ax.text
    (3.38,0.235,'if the clock were the collapse time',ha='right',fontsize=8.5,color='#c0392b')
ax.set_xticks([1,3]); ax.set_xticklabels(['$\\epsilon_{dec}=4$','$\\epsilon_{dec}=8$']); ax.set_ylabel('$G$');
     ax.set_ylim(0,0.6)
ax.legend(fontsize=8.5,frameon=False,loc='upper right',bbox_to_anchor=(1.0,0.98)); ax.set_title('The clock is
    set early: a late collapse does not change $G$',fontsize=11)
fig.tight_layout(); fig.savefig('figs/fig3_clock.png',dpi=220)
print('ok')
```

```
# SCRIPT: PBH-NU-FIGURES-V3
# Figures 4, 5, 6 for the PBH-NU paper. Uses only the numbers and data files listed below.
# Writes fig4_tests.png, fig5_killlines.png, fig6_door.png into figs/. Never touches fig1-fig3.
import json
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from matplotlib.colors import Normalize
from matplotlib.lines import Line2D
from matplotlib.patches import Patch

BASE = '/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad'
PAPER = BASE + '/paper'
MEM = PAPER + '/code/main/memory_curve.json'
WIT = BASE + '/next/witness/witness_prod.json'
OUT = PAPER + '/figs/'
PROTECTED = {'fig1_G.png', 'fig2_chain.png', 'fig3_clock.png'}

plt.rcParams.update({
    'font.family': 'DejaVu Sans',
    'font.size': 9.5,
    'axes.labelsize': 10,
    'xtick.labelsize': 9,
    'ytick.labelsize': 9,
    'legend.fontsize': 8,
    'axes.spines.top': False,
    'axes.spines.right': False,
})

# Okabe-Ito based colour-blind-safe palette
BLUE = '#0072B2'
ORANGE = '#E69F00'
BLACK = '#000000'
GREY = '#7a7a7a'
BAND = '#d6e6f3'
```

```
def save(fig, name, **kw):
    assert name not in PROTECTED, 'refusing to touch an existing figure'
    path = OUT + name
    fig.savefig(path, dpi=300, facecolor='white', **kw)
    plt.close(fig)
    print('wrote', path)


# ---------- data ----------
mc = json.load(open(MEM))


def memory_curve(n):
    pts = []
    for key, val in mc.items():
        nn, ee = key.split('_')
        if int(nn) == n:
            pts.append((float(ee), val['two_epoch']))
    return sorted(pts)


c3 = memory_curve(3)
c6 = memory_curve(6)

# McCormick measured n=3 (history-blind rule)
mc_e = [0.25, 0.5, 1, 2, 4, 8]
mc_G = [0.959, 0.867, 0.683, 0.431, 0.190, 0.063]

# New measured n=6 runs (C, B, D)
new6_e = [1, 2, 3]
new6_G = [0.648, 0.317, 0.154]
new6_err = [0.006, 0.004, 0.006]
new6_lab = ['C', 'B', 'D']

# Broad profile alpha=0.3, n=3
broad_e = [4, 8]
broad_G = [0.181, 0.045]

# ---------- FIG 4 ----------
fig, ax = plt.subplots(figsize=(7.8, 4.6))

ax.plot(mc_e, mc_G, linestyle='--', color=GREY, lw=0.9, zorder=1,
        label='history-blind rule (uses the n=3 curve for any history)')
ax.plot(mc_e, mc_G, 'o', color=BLACK, ms=5.5, zorder=3,
        label='McCormick, n=3 (measured)')

ax.plot([p[0] for p in c3], [p[1] for p in c3], '-', color=BLUE, lw=1.8, zorder=2,
        label='memory model, n=3 (fitted, one weight)')
ax.plot([p[0] for p in c6], [p[1] for p in c6], '-', color=ORANGE, lw=1.8, zorder=2,
        label='memory model, n=6 (frozen, predicted before the runs)')

ax.errorbar(new6_e, new6_G, yerr=new6_err, fmt='D', color=ORANGE, mec=ORANGE,
            ms=6, capsize=3, elinewidth=1.0, zorder=4,
            label="McCormick's code, n=6 (new runs)")
for x, y, lab in zip(new6_e, new6_G, new6_lab):
    ax.annotate(lab, (x, y), xytext=(7, -9), textcoords='offset points',
                fontsize=9, fontweight='bold', color=BLACK)

ax.plot(broad_e, broad_G, 's', mfc='none', mec=BLACK, mew=1.2, ms=6.5, zorder=4,
        label='broad profile, n=3 (new runs)')

ax.set_xscale('log')
ax.set_xlim(0.2, 10)
ax.set_ylim(0, 1)
ticks = [0.2, 0.5, 1, 2, 5, 10]
ax.set_xticks(ticks)
ax.set_xticklabels(['0.2', '0.5', '1', '2', '5', '10'])
```

```
ax.minorticks_off()
ax.set_xlabel(r'decoupling time  $\varepsilon_{\mathrm{dec}}$')
ax.set_ylabel(r'share of the neutrino effect kept, $G$')
ax.legend(loc='upper right', frameon=False, handlelength=2.4)
fig.tight_layout()
save(fig, 'fig4_tests.png')

# ---------- FIG 5 ----------
rows = [
    # label, measured, meas_err, memory, history_blind, annotation
    ('Test B  (n=6, $\\varepsilon_{\\mathrm{dec}}$=2)', 0.317, 0.004, 0.246, 0.431,
     'memory off 0.071 . history-blind off 0.114'),
    ('Test C  (n=6, $\\varepsilon_{\\mathrm{dec}}$=1)', 0.648, 0.006, 0.6625, 0.683,
     'memory off 0.014 . history-blind off 0.035'),
    ('Test D  (n=6, $\\varepsilon_{\\mathrm{dec}}$=3)', 0.154, 0.006, 0.1176, 0.30,
     'memory off 0.037 . history-blind off 0.146 (dies)'),
]
KILL = 0.08

fig, ax = plt.subplots(figsize=(10.5, 3.8))
fig.subplots_adjust(left=0.24, right=0.6, bottom=0.3, top=0.95)
half_h = 0.26
for k, (lab, meas, merr, mem, hb, note) in enumerate(rows):
    y = 2 - k
    ax.fill_betweenx([y - half_h, y + half_h], meas - KILL, meas + KILL,
                     color=BAND, lw=0, zorder=1)
    ax.errorbar([meas], [y], xerr=[merr], fmt='D', color=BLACK, ms=6.5,
                capsize=3, elinewidth=1.0, zorder=5)
    ax.plot([mem], [y], marker='|', color=ORANGE, ms=17, mew=3, ls='none', zorder=4)
    ax.plot([hb], [y], 'o', mfc='none', mec=GREY, mew=1.8, ms=9, ls='none', zorder=4)
    ax.text(1.03, y, note, transform=ax.get_yaxis_transform(),
            va='center', ha='left', fontsize=9, color=BLACK)

ax.set_xlim(0, 0.8)
ax.set_ylim(-0.6, 2.6)
ax.set_yticks([2, 1, 0])
ax.set_yticklabels([r[0] for r in rows], fontsize=9)
ax.set_xlabel('G')
ax.spines['left'].set_visible(False)
ax.tick_params(axis='y', length=0)

handles = [
    Patch(facecolor=BAND, edgecolor='none', label='kill band: +-0.08 around the measurement'),
    Line2D([], [], marker='D', color=BLACK, ls='none', ms=6, label='measured (bar = stated uncertainty)'),
    Line2D([], [], marker='|', color=ORANGE, ls='none', ms=12, mew=3, label='memory-model prediction'),
    Line2D([], [], marker='o', mfc='none', mec=GREY, mew=1.8, ls='none', ms=8,
           label='history-blind prediction'),
]
ax.legend(handles=handles, loc='upper center', bbox_to_anchor=(0.4, -0.2), ncol=2,
          frameon=False)
save(fig, 'fig5_killlines.png')

# ---------- FIG 6 ----------
wit = json.load(open(WIT))
RATIOS = [0.1, 0.25, 0.5, 1, 2]
LS = [1, 2, 3, 4]
FLOOR = -32.0


def height_grid(n):
    H = np.zeros((len(LS), len(RATIOS)))
    for rec in wit:
        if rec['n'] != n:
            continue
        i = int(np.argmin([abs(rec['ratio'] - r) for r in RATIOS]))
        j = LS.index(rec['L'])
        R = rec['res']['4.0']['R_ret']
        H[j, i] = max(np.log10(max(R, 1e-40)) - FLOOR, 0.0)
```

```
    return H


H = {3: height_grid(3), 6: height_grid(6)}
hmax = max(H[3].max(), H[6].max())
ZTOP = max(34.0, np.ceil(hmax) + 2)
norm = Normalize(vmin=0, vmax=ZTOP)
cmap = plt.cm.viridis
WALL_LOG = -3.0
WALL_H = WALL_LOG - FLOOR

fig = plt.figure(figsize=(11.5, 5.4))
axes = []
for k, n in enumerate([3, 6]):
    ax = fig.add_subplot(1, 2, k + 1, projection='3d')
    axes.append(ax)
    Hn = H[n]
    xs, ys, zs, dxs, dys, dzs, cols = [], [], [], [], [], [], []
    for j, L in enumerate(LS):
        for i in range(len(RATIOS)):
            h = Hn[j, i]
            xs.append(i - 0.3)
            ys.append(L - 0.3)
            zs.append(0.0)
            dxs.append(0.6)
            dys.append(0.6)
            dzs.append(h)
            cols.append(cmap(norm(h)))
    ax.bar3d(xs, ys, zs, dxs, dys, dzs, color=cols, shade=True, alpha=1.0,
             edgecolor='none')

    # translucent "witness on" plane at log10 R = -3
    gx, gy = np.meshgrid([-0.6, len(RATIOS) - 0.4], [0.5, 4.5])
    gz = np.full_like(gx, WALL_H, dtype=float)
    ax.plot_surface(gx, gy, gz, color='#555555', alpha=0.22, shade=False,
                    linewidth=0, zorder=0)

    ax.set_xticks(range(len(RATIOS)))
    ax.set_xticklabels(['0.1', '0.25', '0.5', '1', '2'], fontsize=8)
    ax.set_yticks(LS)
    ax.set_yticklabels([str(L) for L in LS], fontsize=8)
    ax.set_zlim(0, ZTOP)
    ax.set_zticks([2, 12, 22, 32])
    ax.set_zticklabels(['-30', '-20', '-10', '0'], fontsize=8)
    ax.set_xlabel('plant time / decoupling time', fontsize=9, labelpad=6)
    ax.set_ylabel('moment that holds the difference, $l$', fontsize=9, labelpad=6)
    ax.set_zlabel(r'$\log_{10} R_{\mathrm{ret}}$', fontsize=9, labelpad=4)
    ax.view_init(elev=28, azim=-32)
    ax.text2D(0.04, 0.9, 'n = %d' % n, transform=ax.transAxes, fontsize=10,
              fontweight='bold', color=BLACK)
    ax.set_box_aspect((1.3, 1.0, 0.95))

fig.subplots_adjust(left=0.03, right=0.97, bottom=0.04, top=0.97, wspace=0.28)
save(fig, 'fig6_door.png', bbox_inches='tight', pad_inches=0.15)
```

```
# SCRIPT: PBH-NU-FIG6-DOOR  witness strength terrain (log10 R_ret at tau=4.0) from witness_prod.json
import json, math, numpy as np, matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from matplotlib import cm
d=json.load(open('/tmp/claude-0/-home-claude/502f5ad8-acd9-59a2-b5cf-d48e7967cc3a/scratchpad/next/witness/
    witness_prod.json'))
ratios=[0.1,0.25,0.5,1,2]; Ls=[1,2,3,4]; FLOOR=-32
fig=plt.figure(figsize=(12,5.2))
for p,n in enumerate((3,6)):
    ax=fig.add_subplot(1,2,p+1,projection='3d')
    for r in d:
        if r['n']!=n: continue
```

```
        i=ratios.index(r['ratio']); j=Ls.index(r['L'])
        z=math.log10(r['res']['4.0']['R_ret'])
        col=cm.viridis((z-FLOOR)/(0-FLOOR))
        ax.bar3d(i-0.35,j-0.35,FLOOR,0.7,0.7,z-FLOOR,color=col,shade=True,edgecolor='k',linewidth=0.3)
    X,Y=np.meshgrid([-0.6,4.6],[-0.6,3.6]); ax.plot_surface(X,Y,np.full_like(X,-3.0),color='red',alpha=0.18)
    ax.set_xticks(range(5)); ax.set_xticklabels([str(x) for x in ratios])
    ax.set_yticks(range(4)); ax.set_yticklabels([str(l) for l in Ls])
    ax.set_zlim(FLOOR,0); ax.set_zticks([-30,-20,-10,-3,0])
    ax.set_xlabel('plant time / decoupling time',labelpad=8); ax.set_ylabel('moment holding the difference, l
     ',labelpad=8); ax.set_zlabel('log$_{10}$ R$_{ret}$ at $\\tau$=4')
    ax.view_init(elev=22,azim=-128)
    ax.text2D(0.03,0.92,'n = %d'%n,transform=ax.transAxes,fontsize=13,weight='bold')
fig.text(0.5,0.02,'red plane: witness on (R$_{ret}$ = 10$^{-3}$)',ha='center',fontsize=10)
plt.tight_layout(); plt.savefig('figs/fig6_door.png',dpi=300,facecolor='white')
```

C Lean statements

The statements below are written in Lean 4 with Mathlib (v4.34.1) in the repository `neutrinos-leave-lean`, with a build that replays every proof in the kernel, audits axioms, and confirms that nine false controls are rejected. A statement counts as machine-checked only once that build passes on a recorded commit; until then it carries the grade of its paper proof. Lean checks the finite moment model and the verdict arithmetic. It does not check McCormick's code, the collapse runs or the numerical witness.

1. `crossLink`: $|2\,\mathrm{Re}(\bar{a}(-ih)b)| \le 2|h||a||b|$; `crossLink_sq`: with $A = |a|^2$, $|a|^2 + |b|^2 \le 1$, the square is at most $4|h|^2A(1-A)$ (Appendix A.6).

2. `twoMomentConnectionEnergy`: $\frac{1}{8}\|[L, L^*]\|_F^2 = \kappa^2 k^2/3$ for the two-moment L.

3. `splitRateEnergy`: $\frac{1}{8}\|[L, L^*]\|_F^2 = k^2\kappa_1^2/3 + \frac{4k^2}{15}(\kappa_2 - \kappa_1)^2$ for the three-moment L.

4. `shearCounterexample`: $R_{00}(1) = 32/37$, $R'_{00}(1) = -5/1369$, $\partial_{\kappa_2} E = 0$ at $\kappa_2 = \kappa_1$; with the complex 3×3 resolvent entry equal to the closed form $2(15\kappa_2 + 17)/(35\kappa_2 + 39)$ for $\kappa_2 > -1$ (Appendix A.7).

5. `scalarCollisionPropagator`: $e^{t(X-\sigma I)} = e^{-\sigma t}e^{tX}$, with generators $X - \sigma(t)I$ commuting pairwise (Appendix A.5).

6. `testB/C/D_verdict`: $G_B = 970/3059$, $G_C = 1983/3059$, $G_D = 472/3059$ against the preregistered kill lines.